STATEMENT OF WORK

FOR A TURN-KEY INSTALLATION OF A SANDING BOOTH

1.0 BACKGROUND:

A new Sanding Booth will provide Tobyhanna Army Depot (TYAD) capability to repair dents, holes, cracks, etc. in composite materials by applying epoxies and glazing compounds, and by performing wet lay-ups using carbon fiber, fiberglass and Kevlar. The Sanding Booth will handle composite material repairs for the depot workload, including radar antennas, radomes, High Mobility Multipurpose Wheeled Vehicle (HMMWV) hoods, transport cases and other similar components. Epoxies and glazing compounds will be applied while wearing proper personal protective equipment, including coveralls, safety goggles and respirators with organic vapor cartridges. The decibel (dBA) sound level of power sanding using a rotary grinder with vacuum attachment is 90-95 dB within the sanding booth enclosure. The desired location of the new booth is Building 30. A diagram of the proposed booth configuration is included under Attachment 1. A dust sample from an existing TYAD sanding booth used for fiberglass sanding and repair was tested at EMSL Analytical, Cinnaminson, NJ for explosivity. The dust sample was determined to be 'explosible' in dust form. The explosion severity result (Kst) is 159 +/- 12%, which is considered Dust Explosion Class St 1, Weak Explosion. The report from the dust explosivity test is included under Attachment 2. Fiberglass repair work performed in the booth involves flammable glues or resins in large volumes; volatile organic compounds (VOCs) are a concern.

2.0 General Scope

2.1 This Statement of Work (SOW) is for the turn-key design, engineering, supply, delivery, installation, assembly, set-up and training for a crossdraft-style sanding booth in an existing building at the Tobyhanna Army Depot (TYAD), Tobyhanna, PA 18466. The Contractor shall supply all materials, supplies, freight, delivery, tools, equipment, consumables, labor, and supervision for the turn-key design, supply, delivery, installation, assembly, set-up and training to install a new crossdraft sanding booth in Building 30 at TYAD in Tobyhanna, PA.

2.2 The new booth needs to incorporate the latest technology improvements. The size of the booth shall be a minimum of 39 feet long by 16 feet wide and 14 feet high inside and be able to accommodate up to four (4) workers. The entire system shall include the crossdraft booth, air filtration system (<u>NOT a dust collection system</u>), front bifold door, rear roll-up door, one regular and one double-wide personnel door, interior LED lighting, central vacuum system, both compressed and breathable air connection points, plus all necessary engineering and design; it must include the site preparation work and installation of the new equipment and any auxiliary equipment. Additional accessories/options, such as fire protection system, fall protection and pull-down hose reels, are described in section 3. (Note: The term "sanding booth" and "booth" used throughout this statement of work refers to all the components that make up a useable sanding booth, such as but not limited to the frame, walls and ceiling panels, doors, duct work, fans, filters and hardware to erect/assemble and provide a useable sanding booth. For brevity, the sanding booth with filters, and fire protection, etc. will be identified as SYSTEM within the remainder of this SOW. When referencing components of the system, the components will be identified individually.

2.3 SITE VISIT: Interested sources are required to perform a site visit. See the solicitation (SF1449) for site visit information. The site visit is for the purpose of the interested sources to consider the layout and floor space of the system within the necessary installation area and to enable the viewing of the delivery, utility, and installation sites. Technical questions must be submitted in writing prior to the due date and time in the SF1449. Should the Contractor fail to identify an obstruction and hindrance to the project that is identifiable from a site visit, the Contractor shall provide a solution to the problem that is to the satisfaction of the Government and at no additional cost to the Government. It shall be the responsibility of the Contractor for all dimensioning of the system including all auxiliary pieces and materials. It shall be the responsibility of the Contractor to bring personal protective equipment (PPE) for the site visit. At a minimum, the PPE needed is protective eyewear, footwear and hearing protection. A Contractor in non-compliance with the PPE requirement may not be permitted on the site visit. The Contractor shall be responsible to verify ALL dimensions. The Contractor shall be responsible to verify that all installed equipment fits within the available installation area. The contractor shall supply all field labor to include supervision and installation labor for, but not limited to, excavation, equipment assembly, concrete and rigging work.

2.4 Within 15 calendar days after award, the Contractor shall coordinate with the Contracting Officer's Representative (COR) the scheduling of a post-award meeting to be held at TYAD at which the Contractor shall discuss the Contractor's action plan and provide a timeline for the completion of the project. It is recommended the Contractor ensure attendance of any sub-contractors that will be used by the Contractor in the execution of this contract.

2.5 Following contract award the Government will supply electronic copies of the existing area in Building 30. The Contractor shall provide a to-scale full-assembly layout drawing package in hardcopy and electronic media (.dwg format) with full dimensioning representing the new system in the building. The drawing package shall be submitted to the COR within 45 calendar days following contract award. The Government will have 15 working days to review the drawing package and to make comments to the Contractor. The Contractor shall begin work and the purchase of all materials only upon receiving approval from the Government of the Contractor-supplied drawing package.

2.6 The Contractor shall not begin site preparation and installation of equipment until given the Notice to Proceed (NTP) by the COR. The Contractor shall have 360 days from NTP to complete all contractual work. Upon receipt of NTP from the COR, the Contractor shall arrive on-site to begin contractual work within 14 calendar days.

2.7 The Contractor shall provide training for sanding booth operator personnel. This training shall focus on basic system operation and features. Training shall also include instruction in maintenance and preventive maintenance (PM) to be performed by sanding booth operator personnel on a daily basis. The training shall be conducted on-site at the system. Training shall be for up to six (6) sanding booth operator personnel. All materials used during training shall become the property of Tobyhanna Army Depot (TYAD).

2.8 The Contractor shall provide training for TYAD maintenance personnel. Maintenance training shall include PM and general maintenance including the lock-out/tag-out procedure to take the system to zero energy state and restore it to full operational capability. Maintenance instruction shall be for up to four (4) maintenance personnel and shall be conducted on-site at the system. All materials used during training shall become the property of TYAD.

2.9 Following completion of all work and prior to Final Acceptance, the Contractor shall furnish to the COR a computer-aided design and drafting (CADD) as-build drawing of the system inside the building. The drawing shall be furnished in PDF format; in .dwg format; and in hardcopy, 22"x34" paper size, in triplicate.

3.0 System Specifications, Characteristics and Features

3.1 The sanding booth shall be a modular self-supporting design and crossdraft type. The sanding booth shall be a minimum of 39 feet long by 16 feet wide and 14 feet high inside. See sketches of floor configuration in Attachment 1. The booth shall be composed of 18 gauge, minimum, steel sidewall panels and 18 gauge, minimum, steel ceiling panels with bolted assembly.

3.2 The sanding booth shall have a minimum of two 12' wide by 13'-6" high (minimum size) product door openings on the short ends of the booth and two personnel doors on one long side of the booth. The front product door on the East end shall be a bifold door, and the rear product door on the West end shall be a roll-up door. The product doors shall have 2" tube frames with single skin sheet metal or similar design for sturdiness with heavy duty hinges for reliable long-lasting service. The personnel door on the North side near the West end shall be a minimum of 2'-6" wide by 6'-8" high with panic latch, and the personnel door on the North side near the East end shall be a double door and shall be capable of opening just one door or both to provide an opening a minimum of 5' wide by 6'-8" high. The personnel access doors shall meet NFPA standards and include vision window and shield.

3.3 The booth shall be a crossdraft-style sanding booth with three-stage filtration plus a fourth stage for volatile organic compounds (VOC) filtration. The cleaned air shall be vented outside through an opening in the exterior wall. The opening through the existing wall and any related ducting for ventilation, as needed, are the responsibility of the contractor.

3.4 The booth exhaust shall be cleaned and scrubbed via a three-stage or similar type air filtration system, plus a fourth stage for VOC filtration. [This is NOT a dust collection system.] The first stage shall consist of roll type media, the second shall be a panel-type filter and the third shall be a bag type filter. The additional fourth stage shall be a VOC filtration system designed to include HEPA Activated Carbon Filtration. Alternative equivalent proposals will be considered. The sanding booth shall employ magnehelic pressure gauges to allow for monitoring of static pressure drop across stages of the filtration system. Magnehelic pressure gauges must be installed to measure the total differential pressure in inches of water column across the exhaust filters.

3.5 The interior of the sanding booth shall contain LED-type lighting fixtures that shall provide 100 foot-candles at 36” above the floor, average, illumination. The light fixtures shall be inside access, 48 inch LED tubes, 4-tube, minimum of 32-watt each tube, T8 ballast 5000k 82 CRI lamps (or equivalent). There shall be a minimum of 28 light fixtures. Light fixtures shall be located on the ceiling and the walls to thoroughly light the inside of the booth and minimize shadows on products and items being processed in the booth. (Fixtures shall have minimum protrusion from ceiling and walls. The lighting fixtures shall be designed and constructed for use in an industrial environment and shall meet all NFPA, OSHA and Local requirements for paint room applications (explosion proof).

3.6 The sanding booth shall be furnished with explosion proof OSHA-approved LED emergency exit lighting and illuminated exit signs.

3.7 The system shall be designed to use either 480V/3 Phase/60 Hz or 208V/3 Phase/60 Hz, and controls may be 120V/1 Phase/60 Hz. Electrical, compressed air and breathable air connectivity will be supplied by the Government up to the exterior connection points on the booth provided by the contractor. It shall be the contractor’s responsibility to install and connect all the electrical controls and utilities to place the sanding booth in service. The Government will provide electrical disconnects per power requirement provided by the contractor. The contractor shall provide to the Government the amount of these and any other utilities required for the operation of the system <u>with the submitted offer</u>. Amps shall be stated in proposal.

3.8 Based on results of fibrous dust explosivity tests (see attachment 2), TYAD Safety Division requires the system to meet National Electrical Code (NEC) Standard for Class 1, Division 1 Environment. All electrical controls shall be minimum of a Type 12 National Electrical Manufacturers Association (NEMA) rated. Control panel shall consist of a UL-508A listed with a single-point connection to main electrical disconnect. Control enclosures shall provide protection from shock and protect against dust, dirt, oil and non-corrosive liquids. If additional disconnects are required, contractor shall identify voltage and amperage requirement 30 days prior to contractor’s need for disconnect.

3.9 The sanding booth shall be designed and constructed to meet all applicable Occupational Safety and Health Administration (OSHA), National Fire Protection Association (NFPA), United Facility Criteria (UFC) and Local codes including, but not limited to, OSHA standards for Ventilation, Occupational Noise Exposure, and Air Contaminants to include any and all options available for additional sound proofing for maximum sound reductions. The booth shall comply with the hazardous location classification system as defined by the National Fire Protection Association (NFPA) 70, National Electric Code (NEC) for NFPA 70 NEC Hazardous Location Class 1, Division 1. <u>The Contractor is responsible to work with a Fire Protection Engineer to ensure that applicable requirements for the booth are identified and met.</u>

3.10 The sanding booth shall follow the current Ventilation Design Criteria standards of the American Conference of Governmental Industrial Hygienists (ACGIH). All make-up air shall come from a clean source <u>outside</u> the building(s) in which the system is located. All air containing dust from any source shall go through the filters and shall be able to be exhausted outside the building. The outside intake and exhaust shall be constructed to prevent precipitation from entering. Make-up air inlet(s) shall be situated and located to avoid contamination from any nearby exhaust air sources. The intake air shall be monitored, at a minimum, for combustible gases, carbon monoxide and carbon dioxide. All outside air intakes and exhausts shall be screened to prevent the entrance of birds and other small animals. No loads, except ducts, shall be placed on or supported by the building structure. All such loads shall be floor supported independent of the building's structural members.

3.11 The system shall be designed and constructed with a <u>water-based</u> fire suppression system. The fire suppression system shall meet United Facility Criteria (UFC) 3-600-01 and NFPA standards. <u>The Contractor is responsible to determine whether the existing water-based fire suppression system has adequate extra capacity for the size of the booth.</u> <u>The Contractor is responsible for making sure the suppression system can handle an extension for the added supply to the new booth.</u> If deemed necessary based on the capacity calculations, the Contractor shall be required to increase the pipe size, increase sprinkler K-factor, provide additional sprinklers, etc. to meet the UFC and NFPA standards. Tobyhanna requires schedule 40 pipe to be used with sprinkler systems. Tobyhanna requires any new or renovated sprinkler piping to be painted red.

3.12 The system shall include explosion proof fire alarm visual and audible notification devices suitable for the conditions within the booth. The sanding booth shall be designed and constructed with fire alarm notification devices that meet UFC and NFPA standards and shall be designed and constructed with fire alarm and alert strobes that meet UFC and NFPA standards. These devices shall be connected to the existing Notifier fire alarm system. Fire Alarm Warning devices shall meet the Class 1 Division 1 classification (Horns, Strobes, Smoke / Heat Detectors).

3.13 All sanding booth framework, wall and ceiling panels shall be painted. The walls shall be white on both the interior and exterior sides.

3.14 Accessories/Options: The sanding booth shall be equipped with the following accessories and options as a minimum:

a. <u>Fall Protection</u> – The booth shall be equipped with two fall protection rails meeting OSHA Fall Protection Regulation 1910.140(c)(13)(i) capable of supporting at least 5,000 pounds (22.2 kN) for each employee attached; or 1910.140(c)(13)(ii) designed, installed, and used, under the supervision of qualified person, as part of a complete personal fall protection system that maintains a safety factor of at least two. Each rail shall be 30' long, running lengthwise on each side of the booth.

b. <u>Pull Down Hose Reels</u> – The booth shall be equipped with three pull down hose reels (PDHRs) for compressed air.

c. <u>Central Vacuum System for use with sanding tools</u> – The booth shall be equipped with a central vacuum collection system with three (3) collection ports (DCPs) along each long side, inside the booth for a total of six (6) ports. It is desired to connect rotary grinder tool and sander hoses to a central vacuum system located outside the booth via the ports along the interior walls of the booth.

d. <u>Utility Niches</u> – Sanding booth shall have a minimum of six each 24" x 24" utility niches in the side walls, (three in one side and three in the other).

e. Air Lines – The booth shall be designed and constructed with a minimum of four (4) breathable air connections and six (6) compressed air connection points both internally and externally.

f. Carbon Monoxide Monitor – The contractor shall supply and install a carbon monoxide monitor equipped with internal explosion proof audible and visual horn / strobe and external display meter / control panel with status lights for breathable air connections.

4.0 Site Preparation

4.1 The Contractor shall be responsible for all site preparation work. The construction site is in an existing building that is occupied by personnel performing work operations. The contractor shall avoid disruption of existing work operations to the extent possible. The contractor shall curtain off the work area or provide other means of preventing or reducing contaminants and noise from impacting adjacent work areas. The Contractor shall provide TYAD Fire and Emergency Services advance notice prior to shutting down or bringing up any fire suppression and protection system such as sprinklers and fire alarms.

4.2 The Contractor shall supply all materials, equipment, labor, supplies, and supervision to install new booth, foundations and associated equipment for the system components per manufacturer specifications.

4.3 The sanding booth is used for fiberglass repair work. Fiberglass (FIBROUS GLASS DUST 65997-17-3 (Fibrous Glass) is considered a nuisance dust by the Occupational Safety and Health Administration (OSHA) and has an exposure standard (see Table 1 below) which needs to be controlled. The following shall be utilized: Enclosing the booth, utilizing HEPA filtered ventilation negative pressure, and utilizing wet methods – all combined to control the hazards while deconstructing the booth. There are additional controls required, both administrative and PPE requirements to be determined in consultation with TYAD Environmental Branch Asbestos Program Manager, Matthew Argust, X6594. The contract shall require a certified air technician to monitor the air standards listed below. Fiberglass waste must be double bagged and goosenecked but can be placed in a regular construction and demolition rolloff. Environmental Branch must inspect remediations prior to the start of work. Point of contact is Devin Zurawski on requirements for air permit modification, X5023.

Table 1 Exposure Standards							
OSHA PEL 8-hour TWA (ST) STEL (C) Ceiling Peak		**NIOSH REL** Up to 10-hour TWA (ST) STEL (C) Ceiling		**ACGIH TLV©** 8-hour TWA (ST) STEL (C) Ceiling		**CAL/OSHA PEL** 8-hour TWA (ST) STEL (C) Ceiling Peak	
PEL-TWA	OSHA Maritime PEL: 15 mg/m^3 as	**REL-TWA**	3 fibers/cm^3 (fibers ≤3.5 µm in	**TLV-TWA**	1 fiber/cc (continuous filament glass fibers, glass wool fibers, rock wool fibers, slag wool fibers and special purpose glass fibers,	**PEL-TWA**	1 fiber/cm^3 [q]

Table 1 Exposure Standards							
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	Fibrous Glass, 8 hr TWA.		diameter, ≥10 µm in length); 5 mg/m^3 (total)		[F]); 5 mg/m^3 (continuous filament glass fibers, inhalable particulate matter]); 0.2 f/cc (refractory ceramic fibers [F]) [1999]		
PEL-STEL		**REL-STEL**		**TLV-STEL**		**PEL-STEL**	
PEL-C		**REL-C**		**TLV-C**		**PEL-C**	
Skin notation	N	**Skin notation**	N	**Skin notation**	N	**Skin notation**	N
Notes: See 29 CFR 1915.1000 Table Z-Shipyards. For General Industry, please see 29 CFR 1910.1000 Table Z-3, Mineral Dusts for Inert or Nuisance Dust. For Construction, see29 CFR 1926.55 Table 1, for Inert or Nuisance Particles.		**Notes:**		**Notes:** See TLV endnote [F].		**Notes:** [q] Fibers per cubic centimeter of air at 25°C and 760 mmHg pressure. To be considered a fiber for this limit the glass particle must be longer than 5µm, have a length to diameter ratio of three or more, and have a diameter less than 3µm. The National Institute for Occupational Safety and Health (NIOSH), Method 7400, Issue 2, August 15, 1994, which is hereby incorporated by reference, shall be used for measuring airborne fiber concentrations.	

4.4 It is highly recommended that demolition be planned for off hours, weekends, or holidays to limit the number of employees potentially exposed. The contractor shall utilize HEPA vacuums for the cleanup process in coordination with TYAD personnel who, prior to booth shutdown, will perform a thorough clean up as they would for a monthly filter change, but not replace the filters. See demolition requirements in sections S3.0 JOB HAZARD ANALYSIS (JHA) / ACTIVITY HAZARD ANALYSIS (AHA) and S11.0 PROTECTION OF PERSONNEL.

4.5 The Contractor shall saw-cut the concrete floor inside the building at the required locations for the booth. The contractor shall use a HEPA Vacuum with a Dust/Silica capture device when drilling or cutting concrete, in conjunction with applicable wet methods with appropriate additional housekeeping measures to ensure that any slurry generated by the wet control method is cleaned up when the work is completed to avoid a secondary dust exposure hazard. The Contractor shall excavate the hole(s) to the required depth(s). The Contractor shall supply and install the manufacturer specified steel reinforcements. The Contractor shall supply, pour and level the concrete using manufacturer’s specified concrete ratings.

4.6 The Contractor shall supply a refuse receptacle for all debris collection and removal. Location of the refuse receptacle will be provided by the Government to the Contractor prior to the start of site preparation work.

4.7 The Contractor shall remove from TYAD all waste and debris generated by site preparation work and installation of the system.

4.8 The Contractor shall clean the work site daily prior to Contractor departure for the day; for weekends; and for holidays.

4.9 For through-wall and roof penetrations, the Contractor shall patch with approved fire sealing material and insulation. Additionally, roof penetrations shall be sealed and weather boots utilized to prevent leaks. All ceiling and wall obstruction removals shall be the responsibility of the Contractor. Prior to removal of wall and ceiling obstructions the Contractor shall notify and obtain approval from the COR.

4.10 The Contractor shall use electric-powered tools and equipment inside the building.

4.11 The Contractor shall supply all Personal Protective Equipment (PPE) for Contractor personnel performing all site preparation work.

5.0 Delivery, Installation, and Assembly

5.1 The Contractor shall use only electric-powered and battery-powered tools and equipment while performing all work inside the building. For Contractor-supplied electrical equipment that requires battery recharging, it shall be the responsibility of the Contractor to supply the battery-charging equipment. Connection of the battery-charging equipment to TYAD power shall be the responsibility of the Contractor.

5.2 The Contractor shall supply all labor, materials, equipment, supervision, and tools to deliver, off-load, locate and place into position, and anchor all system component pieces, materials and supplies necessary for the turn-key assembly and installation of the system.

5.3 The Contractor shall be responsible to uncrate and unpack all supplied equipment. The Contractor shall remove from TYAD all packing, packaging and crating materials. The Contractor shall thoroughly clean all supplied equipment. The Contractor shall supply and install a grounding rod for the system, per manufacturer specification.

5.4 All equipment, tools, materials, supplies and components shall be delivered concurrently. No partial deliveries are acceptable.

5.5 The Contractor may perform all delivery and post-delivery work during normal 1st-shift hours, Monday—Thursday, 8:00 AM—4:00 PM. The Contractor shall not interfere with normal production. The Contractor shall be able to provide delivery on a Saturday. Delivery shall be coordinated by the Contractor with the COR a minimum of 14 calendar days in advance of the delivery at TYAD.

5.6 The Contractor shall supply all labor, tools, equipment, and materials to make all utility connections to all components. The Contractor shall make all inter-component connections. Utilities will be furnished by the Government within 30 feet of the work site.

5.7 The Contractor shall be responsible for anchoring all system components to the floor, per manufacturer's specification, and supplying all items and equipment necessary for this purpose. The Contractor shall be responsible to level the system components per manufacturer's specification. The Contractor shall thoroughly clean the work site prior to daily departure and prior to weekend and holiday departures.

5.8 The Contractor shall furnish all paints, fluids, greases and lubricants required by the manufacturer for the system. The Contractor shall have all paints, fluids, greases and lubricants delivered with the system components and supplies. The Contractor shall furnish Safety Data Sheets (SDSs) for all paints, fluids, greases and lubricants to the COR upon delivery. The Contractor shall remove from TYAD all empty, fully-filled, and partially-filled paint, fluid, grease and lubricant containers upon completion of all work.

5.9 The Contractor shall adhere to all TYAD Environmental, Safety and Fire Standards, codes and regulations. Furnishing of PPE to Contractor personnel shall be the responsibility of the Contractor.

5.10 The Contractor shall furnish a refuse receptacle for waste collection. Location for the refuse receptacle shall be coordinated with the COR prior to its arrival. The Contractor shall remove the refuse receptacle within 3 calendar days of work completion.

5.11 The Contractor shall ensure ease of accessibility to all system components. The Contractor shall ensure 3 feet of clearance, minimum, around all installed components, electrical assemblies, subassemblies and wiring.

5.12 The Contractor shall ensure all wiring is in accordance with National Electric Code (NEC) and OSHA standards. The Contractor shall ensure all installed equipment meets NEC and OSHA standards.

5.13 The Contractor shall be responsible to ensure the system and all components of the system are operating within manufacturer's specifications prior to beginning personnel training.

5.14 The Contractor shall be responsible for start-up and verification of the system.

5.15 The Contractor shall commence installation of the system within 5 working days of equipment and component delivery. Upon initiation of installation, the Contractor shall continue without interruption until all work has been completed. Following receipt of components and equipment at TYAD, the Contractor shall complete assembly; complete installation; complete personnel training; and present all data and documentation to the COR within 120 calendar days.

5.16 The Contractor shall have 360 calendar days from date of receipt of NTP from the COR to complete all contractual work.

6.0 Training

6.1 The Contractor shall provide training for sanding booth operator personnel. This training shall focus on basic system operation and features. Training shall also include instruction in maintenance and preventive maintenance (PM) to be performed by sanding booth operator personnel on a daily basis. The training shall be conducted on-site at the system. Training shall be for up to six (6) sanding booth operator personnel. All materials used during training shall become the property of Tobyhanna Army Depot (TYAD).

6.2 The Contractor shall provide training for TYAD maintenance personnel. Maintenance training shall include PM and general maintenance including the lock-out/tag-out procedure to take the system to zero energy state and restore it to full operational capability. Maintenance instruction shall be for up to four (4) maintenance personnel and shall be conducted on-site at the system. All materials used during training shall become the property of TYAD.

6.3 Personnel training shall commence only after the system and all components of the system have been verified as operating within manufacturer's specifications and shall be conducted on-site by the Contractor or the Contractor's qualified training instructor.

6.4 All Government personnel training shall be conducted in English. All materials and documentation used in the training of Government personnel shall be in American standard English and become the property of TYAD.

7.0 Documentation

7.1 The Contractor shall provide to the COR three (3) hardcopies of the Operation and Maintenance (O&M) Manuals and three (3) hardcopies of the Parts Manuals. The Parts Manuals shall include a list of parts—including consumables—that are recommended by the manufacturers to be kept on-hand at all times. These parts shall be identified by nomenclature, part number, quantity, and price. The O&M Manuals shall contain a complete set of mechanical drawings and diagrams; a complete set of electrical drawings and diagrams including electrical schematics; logic circuitry descriptions; spare parts list; operator and maintenance instructions for operating, testing and maintaining all system components. The O&M Manuals shall contain complete detailed instructions and schedules for system Preventative Maintenance (PM). O&M Manuals shall contain a detailed description of the lock-out/tag-out procedures needed to bring the system and individual components to a zero energy state for maintenance and servicing. Procedures for returning the system and individual components to full operational capability shall also be provided. All documentation shall be provided to the COR prior to final acceptance being given.

7.2 Following installation completion and prior to Final Acceptance, the Contractor shall furnish to the COR a computer-aided design and drafting (CADD) as-built drawing of the installed system inside and outside of Building 10A. The drawing shall be furnished in PDF format; in .dwg format; and in hardcopy, 22"x 34" paper size, in triplicate.

7.3 All documentation shall be written in American standard English.

8.0 General Acceptance

8.1 Upon successful demonstration of the system; completion of operator personnel training and maintenance personnel training; and presentation of all data and documentation to the COR will General Acceptance be granted to the Contractor.

9.0 Final Acceptance

9.1 Final Acceptance will occur upon ACCEPTANCE in Wide Area Workflow (WAWF) of the Contractor's submitted invoice by the COR. Final Acceptance will be granted only after General Acceptance has been given and all terms of the contract have been completed to the satisfaction of the COR.

10.0 Warranty

10.1 The Contractor shall warrant the system, all components, all installation parts and all workmanship for a period of one (1) calendar year following Final Acceptance by the COR. During this warranty period all parts costs, labor costs, transportation and shipping costs, travel costs, tool costs and *per diem* for all Contractor personnel shall be the responsibility of the Contractor. The Contractor field repair technician shall arrive on-site within 72 hours after being contacted by TYAD personnel to make warranty repairs during the warranty period. The Contractor shall supply to the COR Contact Telephone Numbers and Points of Contact (POCs) for use by TYAD personnel for warranty, operations and maintenance questions and support.

10.2 The Contractor shall provide and perform an annual Preventive Maintenance (PM) on the entire system to be included as part of the warranty provided and without additional cost to the Government during the warranty period. The Contractor shall perform the PM within 12 months, but no sooner than 11 months after final acceptance. As a part of this PM the Contractor shall test all components to ensure that all components are functioning per specification. The Contractor shall provide to the COR a complete itemized listing of what was performed as part of the PM, including the means by which the components were tested to ensure their continuing operation within manufacturer's specifications. The Contractor shall provide an itemized parts list with names, part numbers and quantities of the parts that have been replaced as a part of the PM. Date for the PM shall be coordinated by the Contractor with the COR. All replacement parts needed for the PM shall be provided by the Contractor at no charge to the Government and installed by the Contractor at no charge to the Government.

11.0 Safety

11.1 The system and all components shall meet all requirements set forth in OSHA General Industry Standards, current edition. All energy control sources shall have the capability of being locked-out and tagged-out. The Contractor's manuals shall detail the lock-out/tag-out procedures that are required to bring the system and all components to a zero energy state for maintenance and servicing. Procedures for returning the system and all components to full operational capability shall also be provided.

11.2 The machine shall comply with all applicable OSHA and industry safety standards. If equipment is not manufactured in the US and shall bear a CE label of European or other similar standard, under OSHA's NRTL requirements, the product must also have the specific mark of one of the OSHA’s NRTLs recognized to test and certify this particular product.

11.3 Prior to soldering, welding, torch cutting, and all forms of burning and use of open flame, the Contractor shall be responsible to obtain a Burn Permit from the TYAD Fire Department. The Contractor shall use safety screens that offer the required level of protection for the type of work being performed and to provide protection for all personnel in the surrounding area. Soldering, welding, torch cutting, all forms form of burning, require removal of paint prior to operations. Paint must be removed at a minimum of two inches from where soldering, welding, torch cutting, all forms form of burning are performed. Paint may be sanded or ground. All sanding/grinding/drilling equipment is required to have dust/fume capture equipment with HEPA vacuum controls.

11.4 See below the Contractor Safety Standards for TYAD for additional information and requirements.

12.0 Environmental

12.1 The Contractor shall not use fossil-fueled vehicles and fossil-fueled equipment inside the bays and buildings. Only electric vehicles and electric equipment are permissible. If fossil-fueled vehicles or fossil-fueled equipment must be used due to the non-availability of an electric-powered alternative the Contractor shall notify the COR a minimum of 3 working days in advance of its use to permit notification of the appropriate personnel and time for use. Use of fossil-fueled equipment shall be used by the Contractor only during off-shifts, holidays and weekends when no Government personnel will be in the area.

12.2 See below the Standard Specifications for Projects Under the National Environmental Policy Act for additional information and requirements.

13.0 Depot Access

13.1 A background check and approval from TYAD Security Operations Branch is required for all contractor and subcontractor personnel prior to on-site access at TYAD. All persons seeking entrance to TYAD shall submit to and comply with all security standards and requirements in force at the time such persons are seeking entry. All contractors, regardless of resident status or citizenship, will be subject to vehicle search and intense in-processing by TYAD security personnel prior to being granted access to TYAD. This security screening process may be time consuming and access may be delayed or denied. The contractor shall ensure ELTY Form 648-C is completed for all contractor and subcontractor personnel requiring depot access to include warranty services. The TYAD point of contact (POC) will provide ELTY Form 648-C, Request Access to Tobyhanna Army Depot, to the contractor/vendor at least ten days prior to the expected visit date for completion. The contractor/vendor shall return the completed ELTY Form 648-C to the TYAD POC in a timely manner so the same may be submitted to Security for processing no later than seven days prior to the visit. All of the required fields on the form shall be complete and accurate by the contractor/vendor for timely processing. This requirement is inclusive of on-site supervisory or managerial personnel and sub-contractor personnel that the Contractor anticipates will be performing work or visiting on-site. This security screening does not relieve the contractor of any responsibilities to conduct thorough pre-employment background checks and drug screening. Contractor workers will not be granted access to the work site until security screening is completed and access is approved. Any contractor personnel on-site who fail screening will not be permitted further access to TYAD. See "Access and General Protection/Security Policy and Procedures" below.

13.2 Please submit the completed ELTY Forms 648-C form(s) to the COR or POC.

14.0 TYAD SECURITY REQUIREMENTS FOR THE CONTRACTOR

ANTI-TERRORISM OPERATIONS SECURITY (AT OPSEC) REQUIREMENTS:

Security Operations (SECOPS) Requirements to be Inserted into All Contracts (Updated July 2024)

All Tobyhanna Army Depot (TYAD) contracts and other acquisition-related documents must ensure privacy and security controls follow this information, and that contractors and service providers protect Privacy Act information in the same way the organization adhering to the Federal Acquisition Regulations (FAR) Privacy Act provisions (Subparts 24.1 and 24.2) and include the specified contract clauses (Parts 52.224-1 and 52.224-2), as appropriate, to ensure that personal information is protected as mandated.

In addition to the changes authorized by the clause of this contract; should Force Protection Condition (FPCON) at the installation change, the Government may require changes in contractor security matters and/or processes.

FPCON impact on work levels.

Current FPCON: Bravo

FPCON Charlie: Applies when a hostile incident occurs within the CDR's area of interest (AOI), or intel is received indicating a hostile act or targeting against DoD elements, PAX, or facilities is likely.

FPCON Delta: Applies when a terrorist attack or hostile act has occurred or is anticipated against specific installations or operating areas.

(Please annotate with an X which option may apply.

If option 1 applies, indicate which contracted services will be affected in the brackets)

1. __X__ During FPCONs Charlie and Delta, all services are discontinued. All services will resume when the FPCON level is reduced to level Bravo or lower.
2. _____This contract and its employees are considered mission essential. Therefore, all contractor employees are required to report for duty and remain on duty during declared emergencies and/or elevated FPCON levels unless otherwise directed by the contacting officer via the appropriate COR.

Unscheduled gate closures by the Security Police may occur at any time causing all personnel entering or exiting a closed installation to experience a delay. This cannot be predicted or prevented. Contractors are not compensated for unexpected closures or delays.

Vehicles operated by contractor personnel are subject to search pursuant to applicable regulations. Any moving violation of any applicable motor vehicle regulation may result in the termination of the contractor employee's installation driving privileges.

The contractor's employees shall become familiar with and obey the regulations of the installation, including fire, traffic, safety and security regulations while on the installation. Contractor employees should only enter restricted areas when required to do so and only upon prior approval. All contractor employees shall carry proper identification with them at all times. The contractor shall ensure compliance with all regulations and orders of the installation which may affect performance.

(*If Required*) The contractor shall provide support during contingencies, exercises, heightened operations, and adverse weather or security closures in the accomplishment of performance requirements. From time to time, the Base Commander may decide to close all or part of a base in response to an unforeseen emergency or similar occurrence. Such emergencies include adverse weather such as snow, or ice, "an act of God," such as tornado or earthquake, or a base disaster such as a gas leak or fire.

Emergency Communications and Services.

If assigned to TYAD for seven-days or longer, contractors will enroll in the ALERT! Mass Warning Notification System (MWNS) at https://alertservices.csd.disa.mil/.

Alert! MWNS is NOT optional - it is a Headquarters Department of the Army (HQDA) requirement to be a registered user.

Due to the life-safety implications of the information being relayed and the requirement to provide immediate alerts and warnings, personnel assigned to TYAD must ensure that their personal contact information, including after-duty hours contact information, as appropriate (e.g., personal cellular phone numbers or landline phone numbers), e-mail addresses, home address, etc., are entered into the system and regularly updated or verified every 90 days to remain current and accurate.

- A Common Access Card (CAC) is required to register for ALERT!

- If a CAC has not been deemed necessary for the contractor (see below paragraph: *Common Access Card (CAC),* COR will send the following information to usarmy.tyad.tyad.mbx.secops-at@army.mil to manually input personnel data into ALERT!
- Full Name
- Cell Phone
- Is SMS ok for alerts? Yes or No
- Email
- Contractor: For what Cost Center
- Name of COR
- Expected length of contract

In the event of a life-threatening emergency or natural disaster, COR will account for assigned contractors. COR will send personnel accountability report to Emergency Operations Center (EOC).

Base closure announcements will be disseminated via ALERT! MWNS.

Antiterrorism (AT) Level I Training. All contractor employees, to include subcontractor employees, requiring access to Army installations, facilities, and controlled access areas shall complete AT Level I awareness training prior to contract report date. This training is required for any additional or new contractor employees, who start after that period. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee, to the COR/POC within **10 calendar days** after completion of training by all employees and subcontractor personnel. AT level I awareness training is available at the following website: https://jko.jten.mil course# JS-US007 for their training. Completion of contractor employee training will be documented on *ELTY form 583*, TYAD On-Post Training Record, or contractor equivalent. As applicable, contractor employees must complete annual AT awareness training as it pertains the length of the contract.

iWATCH Army Training. This locally developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees commencing performance with the results reported to the COR NLT 60 calendar days after contract award. Contractors and all associated sub-contractors will be briefed by the TYAD Antiterrorism/Force Protection officer. It is recommended this training be conducted as part of an initial entry briefing or unit/organization newcomer's briefings. Send training requests to usarmy.tyad.tyad.mbx.secops-at@army.mil, completion of contractor employee training will be documented on *ELTY form 583*, TYAD On-Post Training Record or contractor equivalent.

Access and General Protection/Security Policy and Procedures. Contractor and all associated sub-contractor employees shall comply with applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative). The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by TYAD Law Enforcement. Contractor workforce must comply with all personal identity verification requirements as directed by DoD, HQDA, and/or local policy.

Contractor and all associated sub-contractor employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB) (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by the government representative), or, at OCONUS locations, in accordance with status of forces agreements and other theater regulations.

A background check and approval from Tobyhanna Army Depot (TYAD) Law Enforcement is required for all contractor and subcontractor personnel prior to on-site access at TYAD. All persons seeking entrance to TYAD shall submit to and comply with all security standards and requirements in force at the time such persons are seeking entry. All contractors, regardless of resident status or citizenship, will be subject to vehicle search and intense in-processing by TYAD security personnel prior to being granted access to TYAD. This security screening process may be time consuming, and access may be delayed or denied. The contractor shall ensure ELTY Form 648-C is completed for all contractor and subcontractor personnel requiring depot access to include warranty services. The TYAD point of contact (POC) will provide *ELTY Form 648-C, “Request Access to Tobyhanna Army Depo*t” to the contractor/vendor at least **ten days** prior to the expected visit date for completion. The contractor/vendor shall return the completed *ELTY Form 648-C* to the TYAD POC in a timely manner so the same may be submitted to Security for processing no later than **seven days** prior to the visit. All the required fields on the form shall be complete and accurate by the contractor/vendor for timely processing. This requirement is inclusive of on-site supervisory or managerial personnel and sub-contractor personnel that the contractor anticipates will be performing work or visiting on-site. This security screening does not relieve the contractor of any responsibilities to conduct thorough pre-employment background checks and drug screening. Contractor workers will not be granted access to the work site until security screening is completed and access is approved. Any contractor personnel on-site who fail screening will not be permitted further access to TYAD.

Submit the completed *ELTY Forms 648-C* form(s) to the COR or POC. Ensure contracts include the provisions that check for the possibility of and prevent undocumented workers for inclusion in contracted work related to Army missions.

The company will ensure that its employees entering Army-controlled installations or facilities have obtained access badges and passes in accordance with facility regulations and that these badges and passes are obtained in advance so as not to delay the accomplishment of contracted services.

Common Access Card (CAC): *[If applicable.]* Before CAC issuance, the contractor employee requires, at a minimum, a favorably adjudicated National Agency Check with Inquiries (NACI) or an equivalent or higher investigation in accordance with Army Directive 2014-05. The contractor employee will be issued a CAC only if duties involve one of the following: (1) Both physical access to a DoD facility and access, via logon, to DoD networks on-site or remotely; (2) Remote access, via logon, to a DoD network using DoD-approved remote access procedures; or (3) Physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. At the discretion of the sponsoring activity, an initial CAC may be issued based on a favorable review of the FBI fingerprint check and a successfully scheduled NACI at the Office of Personnel Management.

Contractors shall be identified with Government issued identification card (e.g., Common Access Card (CAC) or unit specified identification card). When required, contractor personnel shall comply with local security policies to wear their identification card in a standardized manner, clearly visible attached to the torso of the exterior garment above the belt and below the shoulders except when in use (i.e., inserted in a computer CAC reader) or when in controlled areas requiring other credentials as the primary method of identification. To access any Government base and certain facilities, the contractor shall present required identification card upon demand. Upon exit from Government facilities, the contractor shall conceal their credentials (CAC or other credentials) from plain view. The contractor and Contractor Security Manager/Officer shall coordinate with the COR for Government credential issues.

For contractors that do not require CAC, but require access to a DoD facility or installation: Contractor and all associated subcontractor employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (Army Directive 2014-05/AR 190-13); applicable installation, facility and area commander installation and facility access and local security policies and procedures (provided by Government representative); or, at OCONUS locations, in accordance with status-of-forces agreements and other theater regulations.

Collection of Badges: The Contractor shall account for all forms of Government-provided identification issued to the Contractor employees in connection with performance under this contract. The Contractor shall return such identification to the issuing agency at the earliest of any of the following, unless otherwise determined by the Government:

1. When no longer needed for contract performance
2. Upon completion of the Contractor employee's employment
3. Upon contract completion or termination

The Contracting Officer may delay final payment under a contract if the Contractor fails to comply with these requirements. The Contractor shall insert the substance of this clause, including this paragraph, in all subcontracts when the subcontractor's employees are required to have routine physical access to a federally controlled facility and/or routine access to a federally controlled information system. It shall be the responsibility of the prime Contractor to return such identification to the issuing agency in accordance with the terms set forth in of this section, unless otherwise approved in writing by the Contracting Officer. The company will return all issued U.S. Government Common Access Cards, installation badges, and/or access passes to the COR when the contract is completed or when a contractor employee no longer requires access to the installation or facility. Contractor personnel will obtain a vehicle pass for access to the military installation and Common Access Cards (CAC) for computer access, if applicable.

Security Clearances. *[If applicable.]* Performance of work will require access to classified information or equipment IAW the DD Form 254, Contract Security Classification Specification, provided as an attachment. Contractor shall comply with FAR 52.204-2, Security Requirements. This clause involves access to information classified “Confidential,” “Secret,” or “Top Secret” and requires contractors to comply with— (1) The Security Agreement (DD Form 441), including the National Industrial Security Program Operating Manual (DoD 5220.22-M); (2) any revisions to DOD 5220.22-M, notice of which has been furnished to the contractor. If subsequent to the date of this contract, the security classification or security requirements under this contract are changed by the Government and if the changes cause an increase or decrease in security costs or otherwise affect any other term or condition of this contract, the contract shall be subject to an equitable adjustment as if the changes were directed under the Changes clause of this contract.

The Contractor agrees to insert terms that conform substantially to the language of this clause but excluding any reference to the Changes clause of this contract, in all subcontracts under this contract that involve access to classified information. Contractor personnel performing IT sensitive duties are subject to investigative and assignment requirements IAW AR 25-2, AR 380-67, DoD 8570.0 and affiliated regulations. Army regulation available at www.apd.army.mil.

Threat Awareness and Reporting Program (TARP). *[If applicable.]* For all contractors with security clearances, per AR 381-12, TARP contractor employees must receive initial and annual TARP training by a CI Agent or other trainer as specified.

Insider Threat Program. The contractor will establish and maintain an insider threat program to gather, integrate, and report relevant and available information indicative of a potential or actual insider threat, consistent with E.O. 13587 and Presidential Memorandum “National Insider Threat Policy and Minimum Standards for Executive Branch Insider Threat Programs.”

COMSEC/IT Security. All communications with DoD organizations are subject to communications security (COMSEC) review. All telephone communications networks are continually subject to intercept by unfriendly intelligence organizations. DoD has authorized the military departments to conduct COMSEC monitoring and recording of telephone calls originating from, or terminating at, DoD organizations. Therefore, the contractor is advised that any time contractor place or receive a call they are subject to

COMSEC procedures. The contractor shall ensure wide and frequent dissemination of the above information to all employees dealing with DOD information. The contractor shall abide by all Government regulations concerning the authorized use of the Government's computer network, including the restriction against using the network to recruit Government personnel or advertise job openings.

National Defense Authorization Act Section 889 Compliance: DoD, GSA, and NASA have issued an interim rule amending the Federal Acquisition Regulation (FAR) to implement section 889(a)(1)(B) of the John S. McCain National Defense Authorization Act (NDAA) for Fiscal Year (FY) 2019 (Pub. L. 115-232). Section 889(a)(1)(B) prohibits executive agencies from entering into, extending, or renewing a contract with an entity that uses any equipment, system, or service that uses covered telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system, on or after August 13, 2020, unless an exception applies, or a waiver is granted. See solicitation provision 52.204-24 and clause 52.204-25.

Operations Security (OPSEC). OPSEC shall be considered across the entire spectrum of DoD missions, functions, programs, and activities. OPSEC is applicable to all personnel, missions, and supporting activities on a daily basis. For OPSEC to be effective, all Army personnel (Soldiers, Family members, DA civilians, and contractors) must be aware of OPSEC and understand how it compliments traditional security programs. OPSEC principles and tactics protect not just organizations, but also individuals, their families, and other loved ones.

Personnel who knowingly, willfully, or negligently fail to protect critical information from unauthorized disclosure may be subject to administrative, disciplinary, contractual, or criminal action.

OPSEC awareness and execution is crucial to Army Success. The risk of exposure to critical or classified information is mitigated by providing OPSEC training.

Contracts that Require an OPSEC Standing Operating Procedure/Plan.
[If applicable.] The contractor shall develop an OPSEC Standing Operating Procedure (SOP) or OPSEC Plan within 90 calendar days of contract award, to be reviewed and approved by the responsible Government OPSEC officer, per AR 530-1, Operations Security. This SOP/Plan will include the organization's critical information, why it needs to be protected, where it is located, who is responsible for it, and how to protect it. In addition, the contractor shall identify an individual who will be an OPSEC Coordinator. The contractor will ensure this individual becomes OPSEC Level II certified as per AR 530-1.

Operations Security Training. In accordance with AR 530-1, Operations Security, (4-2-a) OPSEC Level I certification training. The target audience for Level I is all personnel (which includes Soldiers, civilians, and contractors). Level I training is composed of both **initial** and **continual** (annual) awareness training and both shall be completed prior to contract report date.

1. **Initial** OPSEC awareness training. All newly assigned personnel **within the first 30 days** of arrival in the organization (this includes accessions and initial entry programs) must receive initial training. It is recommended this training be conducted as part of an initial entry briefing or unit/organization newcomer's briefings. This training is provided by TYADs OPSEC officer and can be conducted via distance learning, providing all necessary objectives are met.

Initial training focuses on TYAD-specific critical information and countermeasures. **Contractors must email TYADs OPSEC Officer to schedule OPSEC Level I Initial training (in person or via MS Teams). Initial training may not be "read and initial".**

2. **Continuous** (annual) OPSEC awareness training. OPSEC awareness training must be continually provided to the workforce, reemphasizing the importance of sound OPSEC practices.

OPSEC Level I Continuous (annual) training is available at the following website: https://securityawareness.usalearning.gov/opsec/index.htm

The contractor shall submit OPSEC Level I Continual (annual) certificates of completion for each affected contractor employee and subcontractor employee, to the COR (before the contract report date) within 10 calendar days after completion of training. Completion of training may be documented on an ELTY 583, TYAD On-Post Training Record or contractor equivalent.

OPSEC Countermeasures. The specific OPSEC countermeasures that follow are intended to eliminate or mitigate vulnerabilities. Complying with established security policies and procedures also supports the overall OPSEC objective. All personnel (Soldiers, civilians, and contractors) at every level are required to protected critical information that could potentially be exploited by adversaries. The following OPSEC countermeasures support the protection of critical information identified in TYADs OPSEC Level I Initial training, TYADs Critical Information List, and they are required to be implemented by everyone in all day-to-day activities.

1. Complete mandatory training, practice OPSEC, and implement TYAD OPSEC countermeasures to protect critical information (CI) and other sensitive unclassified information in the execution of military operations performed or supported by the contractor in support of the mission. Protection of CI will include the adherence to and execution of countermeasures which the contractor initiates or as provided by TYAD, for CI on or related to the SOW/PWS.

2. Contractor personnel shall not discuss government operations in public or over unprotected or unencrypted communications. Official business and controlled unclassified information (CUI) may only be transmitted as directed in the SOW/PWS. CUI must be marked and protected according to DoD and TYAD regulations. Do not take CUI off the installation.

3. Because observation of events, operations, physical changes, etc. may reveal National Security information, specific restrictions are needed to preclude unintentional release of this information to unauthorized parties. (Unauthorized disclosure and transfer of National Security Information is punishable under 18 USC § 793.) Therefore, contractor personnel shall **not** disclose to unauthorized third parties, post to company websites, unofficial sites, publications, newsletters, or other media (including Social Networking sites) any images, data or information, or observed events that reveal critical/sensitive government information about operations, personnel, equipment, including, but not limited to: tactics, techniques and procedures, production or work schedules, any visible or concealed modifications, upgrades, additions to vessels, aircraft, or weapons or equipment; increases, change, or decreases in work/deployment frequency or government personnel, vehicle, vessel or aircraft movements; specialized equipment orders, deliveries, shipments, etc.,

a. A documented OPSEC reviews will be conducted of all information prior to being released to the public regardless of format. OPSEC Reviews must be completed and documented by an OPSEC Level II certified individual.

b. When in doubt, company press releases related to this contract should be coordinated through the Contracting Officer Representative (COR) or Technical Point of Contact, as applicable.

4. Unauthorized disclosures and attempts to solicit classified or critical information by unauthorized third parties or others not affiliated with this contract shall be reported to the TYADs Security Operations Branch, TYADs Law Enforcement Branch, contract point of contact, and your company Facility Security Officer and/or the Defense Security Service. Non-Disclosure requirements remain in effect during the duration of this contract and indefinitely thereafter.

5. Government issued badges, identification shall be removed and/or concealed from plain sight when off station and shall not be left in vehicles or unprotected. Badges and passes may not be duplicated or copied or loaned to others. Lost or stolen identification badges, vehicle passes etc. will be immediately reported to the installation Security Office.

6. It is strongly recommended the contractor mark and protect related internal production schedules, deliverables, inventories, shortages, and identified vulnerabilities related to production of government material. Internal company markings e.g., Business Sensitive, etc., are appropriate for identifying the aforementioned as sensitive information. Specific Government-provided information, drawings etc., will be protected in accordance with guidance in applicable paragraphs of the SOW.

7. All government information must be properly destroyed at contract termination or returned to the government at the government's discretion.

Information Security (INFOSEC)

Contractor personnel must comply with local security requirements for entry and exit control for personnel and property at the Government leased facility or any Government facility where work is being performed.

Contractor employees will be required to comply with all Government security regulations and requirements. Initial and periodic security training and briefings will be required. Failure to comply with security requirements can cause for removal and the Contractor will not be permitted to provide service on this contract.

The Contractor shall not divulge any information about DoD files, data processing activities or functions, user identifications, passwords, or any other knowledge that may be gained, to anyone who is not authorized to have access to such information. The Contractor shall observe and comply with the security provisions in effect at the Government leased facility or any other Government facilities where work is being performed. Identification shall be worn and displayed as required.

Safeguarding Controlled Unclassified Information (CUI)

CUI is Government-created or owned UNCLASSIFIED information that allows for or requires safeguarding and dissemination controls in accordance with laws, regulations, or Government-wide policies. It is sensitive or critical information that does not meet the criteria for classification but must still be protected. CUI policy provides a uniform marking system across the Federal Government that replaces a variety of agency-specific marking, such as FOUO, LES, SBU, etc. CUI markings alert recipients that special handling may be required to comply with law, regulation, or Government-wide policy. The DoD CUI Registry will give you information on every category to include a description of the category, required markings, authorities and DoD policies, and examples. **Not every category or authority listed in the Registry will be applicable to DoD.** More specific information about CUI can be found on the DoD CUI Program website: https://www.dodcui.mil/Home/Training/

The contractor (and/or any subcontractor) must comply with Executive Order 13556, Controlled Unclassified Information, (implemented at 32 CFR, part 2002) when handling CUI. 32 CFR 2002.4(aa) as implemented the term “handling” refers to “…any use of CUI, including but not limited to marking, safeguarding, transporting, disseminating, re-using, and disposing of the information.” 81 Fed. Reg. 63323. All sensitive information that has been identified as CUI by a regulation or statute, handled by this solicitation/contract, shall be:

a. Marked appropriately;
b. Disclosed to authorized personnel on a “Need-To-Know” basis;
c. Protected in accordance with NIST SP 800-53, Rev. 4 Security and Privacy controls for Federal Information Systems and Organizations applicable baseline if handled by a Contractor system operated on behalf of the agency, or NIST SP 800-171, Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations if handled by internal Contractor system; and

d. Returned to TYAD control, destroyed when no longer needed, or held until otherwise directed. Destruction of information and/or data shall be accomplished in accordance with NIST SP 800-88, Guidelines for Media Sanitization.

Safeguarding Sensitive Information: For security purposes, information is or may be sensitive because it requires security to protect its confidentiality, integrity, and/or availability. The contractor (and/or any subcontractor) shall protect all government information that is or may be sensitive in accordance with FISMA by securing it with a FIPS 140-2 validated solution.

Confidentiality, Integrity, Availability, and Nondisclosure of Information: Any information provided to the contractor (and/or any subcontractor) by TYAD or collected by the contractor on behalf of TYAD shall be used only for the purpose of carrying out the provisions of this contract and shall not be disclosed or made known in any manner to any persons except as may be necessary in the performance of the contract. The contractor assumes responsibility for protection of the confidentiality, integrity, and availability of Government records and shall ensure that all work performed by its employees and subcontractors shall be under the supervision of the contractor. Each contractor employee or any of its subcontractors at any level to whom any TYAD records may be made available or disclosed shall be notified in writing by the contractor that information disclosed to such employee or subcontractor can be used only for that purpose and to the extent authorized herein. The confidentiality, integrity, and availability of such information shall be protected in accordance with TYAD policies and instructions. Unauthorized disclosure of information will be subject to the TYAD sanction policies and/or governed by the following laws and regulations:

- 18 U.S.C. 641 (Criminal Code: Public Money, Property or Records);
- 18 U.S.C. 1905 (Criminal Code: Disclosure of Confidential Information); and
- 44 U.S.C. Chapter 35, Subchapter I (Paperwork Reduction Act).
- 18 U.S.C. 1030 The Computer Fraud and Abuse Act (CFAA)
- 44 U.S.C. 3301 Definition of Records

Government Access for Security Assessment: In addition to the Inspection Clause in the contract, the contractor (and/or any subcontractor) shall afford the Government access to the contractor's facilities, installations, operations, documentation, information systems, and personnel used in performance of this contract to the extent required to carry out a program of security assessment (to include vulnerability testing), investigation, and audit to safeguard Security and Privacy Requirements for Information Technology Procurements against threats and hazards to the confidentiality, integrity, and availability of federal data or to the protection of information systems operated on behalf of TYAD, including but are not limited to:

a. At any tier handling or accessing information, consent to and allow the Government, or an independent third party working at the Government's direction, without notice at any time during a weekday during regular business hours contractor local time, to access contractor and subcontractor installations, facilities, infrastructure, data centers, equipment (including but not limited to all servers, computing devices, and portable media), operations, documentation (whether in electronic, paper, or other forms), databases, and personnel which are used in performance of the contract. The purpose of the access is to facilitate performance inspections and reviews, security and compliance audits, and law enforcement investigations. For security audits, the audit may include but not be limited to such items as buffer overflows, open ports, unnecessary services, lack of user input filtering, cross site scripting vulnerabilities, Structured Query Language (SQL) injection vulnerabilities, and any other known vulnerabilities.

b. At any tier handling or accessing protected information, fully cooperate with all audits, inspections, investigations, forensic analysis, or other reviews or requirements needed to carry out requirements presented in applicable law or policy. Beyond providing access, full cooperation also includes, but is not limited to, disclosure to investigators of information sufficient to identify the nature and extent of any criminal or fraudulent activity and the individuals responsible for that activity. It includes timely and complete production of requested data, metadata, information, and records relevant to any inspection, audit, investigation, or review, and making employees of the contractor available for interview by inspectors, auditors, and investigators upon request. Full cooperation also includes allowing the Government to make reproductions or copies of information and equipment, including, if necessary, collecting a machine or system image capture.

c. Cooperate with inspections, audits, investigations, and reviews.

Physical Security: The Contractor shall be responsible for safeguarding all government equipment, information and property provided for Contractor use. At the close of each work period, government facilities, equipment and materials shall be secured.

Key Control. *[If applicable.]* The Contractor shall establish and implement methods of making sure all keys/key cards issued to the Contractor by the Government are not lost or misplaced and are not used by unauthorized persons. ***NOTE:*** All references to keys include key cards. No keys issued to the Contractor by the Government shall be duplicated. The Contractor shall develop procedures covering key control that shall be included in the Quality Control Plan. Such procedures shall include turn-in of any issued keys by personnel who no longer require access to locked areas. The Contractor shall immediately report any occurrences of lost or duplicate keys/key cards to the KO.

In the event keys, other than master keys, are lost and/or duplicated, the Contractor shall, upon direction of the KO, re-key or replace the affected lock or locks; however, the Government, at its option, may replace the affected lock or locks or perform re-keying. When the replacement of locks or re-keying is performed by the Government, the total cost of re-keying or the replacement of the locks or locks shall be deducted from the monthly payment due to the Contractor. In the event a master key is lost or duplicated, all locks and keys for that system shall be replaces by the Government and the total cost deducted from the monthly payment due to the Contractor.

The Contractor shall prohibit the use of Government issued keys/key cards by any persons other than the Contractor's employees. The Contractor shall prohibit the opening of locked areas by Contractor employees to permit entrance of persons other than Contractor employees engaged in the performance of assigned work in those areas or personnel authorized entrance by the KO.

Lock Combination. *[If applicable.]* The Contractor shall establish and implement methods of ensuring that all lock combinations are not revealed to unauthorized persons. The Contractor shall ensure that lock combinations are changed when personnel having access to the combinations no longer have a need to know such combinations. These procedures shall be included in the Contractor's Quality Control Plan.

All security training certificates shall be provided to the Contracting Officer's Representative (COR)/Point of Contact (POC) and the Contract Specialist/Purchasing Agent.

15.0 TOBYHANNA ARMY DEPOT ON-SITE CONTRACTOR WORKFORCE AND VISITOR RESTRICTIONS

1. NON-RESIDENT/NON-IMMIGRANT ALIENS

a. All non-resident/non-immigrant aliens must have approval prior to being permitted access to Tobyhanna Army Depot (TYAD). Such approval must be obtained by requesting access through the following: apply at the alien-resident's embassy; proceed to the U.S. Embassy; proceed to the Department of the Army; proceed to Army Materiel Command; proceed to U.S. Communications-Electronics Command; proceed to TYAD.

b. All non-resident/non-immigrant aliens granted access to TYAD are required to be escorted by Government personnel. One Government escort can accommodate a maximum of two non-resident/non-immigrant aliens.

c. Due to limited availability of Government personnel, contractors shall not be permitted to employ non-resident/non-immigrant aliens as part of the contractor's on-site workforce. Limited exceptions to this restriction may be considered on a case-by-case basis. Such exceptions shall only be considered where a specialized skill or trade is not otherwise available, and even under such circumstances would only be considered for very limited duration, e.g., a few hours/days. However, even if limited exception is considered, access may still be denied. Contractors whose personnel are either denied entry to TYAD or permitted limited entry to TYAD due to the security requirements pertaining to non-resident aliens/non-immigrants are not relieved of their obligation to provide the required contract performance and must do so at no additional cost to the Government. Any contractor failing to perform or that fails to perform in a timely manner under such circumstances may be terminated for default or other contractual remedies as appropriate.

d. Due to limited availability of Government personnel, visitors requiring escort may not be permitted access, or access may be limited to a certain time and duration.

2. FOR THE PURPOSE OF FORMAL ESCORTED SITE VISITS

Resident aliens (immigrants) in possession of a valid Form I-551, Alien Registration Receipt Card ("Green Card"), are not required to obtain approval as set forth in paragraph 1.a. above (but see paragraph 5, below). Such individuals must have proper identification in addition to a valid form I-551 in their physical possession at all times while at TYAD. Individuals lacking proper identification and valid form I-555 will be denied access to TYAD.

3. SECURITY SEARCHES, IN-PROCESSING, AND SECURITY SCREENING

All persons seeking entrance to TYAD must submit to and comply with all security standards and requirements in force at the time such persons are seeking entry. All visitors and contractors, regardless of resident status or citizenship, will be subject to vehicle search and intense in-processing by TYAD Security personnel prior to being granted access to TYAD. This security screening process may be time-consuming and access may be delayed or denied.

Any Contractors who will be working on site, company representatives who will be visiting periodically, and any companies or contractors requesting or requiring TYAD badges for other business reasons must submit a completed TYAD form 648-C to their contract or depot point of contact. The completed form must be submitted at least five (5) workdays in advance of on-site performance for each employee intended for onsite performance or five (5) workdays in advance of visits or other business at the depot. Any contractor employee (s) already on site who have not been screened, will be subject to security screening. This requirement is inclusive of on-site supervisory or managerial personnel and sub-contractor personnel that the Contractor anticipates will be performing work or visiting on-site. This security screening does not relieve the contractor of any responsibilities to conduct thorough pre-employment background checks and drug screening. Contractor workers will not be granted access to the work site until security screening is completed and access is approved. Any contractor personnel on site who fail screening will not be permitted further access to TYAD.

4. SUBMISSION OF BIDS OR PROPOSALS

Bidders/offerors who hand carry bids or quotes do so at their own risk. Bidders/offerors are solely responsible for the timely submission of bids /proposals/quotations, any delays security measures notwithstanding.

5. CONTRACTOR ON-SITE WORKFORCE - ADDITIONAL SECURITY REQUIREMENTS

A. This section is in addition to the requirements above regarding non-resident aliens (non-immigrants) for on-site performance. Prior to the commencement of performance under the contract and within seven (7) workdays of contract award, the Contractor shall submit to the Chief, Security Division, Tobyhanna Army Depot, a roster of all contractor personnel, inclusive of on-site supervisory or managerial personnel and sub-contractor personnel, that the Contractor anticipates will be performing work on-site. The roster shall indicate which individuals are U.S. citizens and which are resident aliens (immigrants). The following documentation shall accompany the roster for each individual named on the roster as a resident alien (immigrant):

a) A copy of a verifiable form of identification, such as a driver's license or a passport; and
b) A copy of a valid Department of Justice Immigration and Naturalization Service Form I-551, Alien Registration Receipt Card ("Green Card").

Contractors who fail to identify any and all resident aliens (immigrants) who will be used for on-site performance and who will seek access to Tobyhanna Army Depot as a worker for or through the contractor, inclusive of managerial and subcontractor personnel, may be subject to civil and criminal penalties and sanctions as well as contract remedies.

B. Within five (5) workdays of the submission of the roster, documentation, and TYAD form 648-C, the Government will have completed the security check and the Contractor shall be notified whether all listed personnel will be permitted to work on-site. However, actual access to Tobyhanna Army Depot by the roster personnel will remain contingent upon such resident alien (immigrant) individuals presenting two (2) forms of identification as they process into the Depot through the Security building: a valid Form I-551, Alien Registration Receipt Card ("Green Card") in addition to at least one other verifiable form of identification.

C. The roster that the Contractor submits is amendable. However, up to five (5) workdays will be required to perform a security check on any personnel added to the roster. The Contractor is urged to include contingency or "back-up" personnel in the original roster in order to avoid delays due to roster amendment.

D. Contractors whose personnel are denied entry to TYAD due to the security requirements are not relieved of their obligation to provide the required contract performance and must do so at no additional cost to the Government. Any contractor failing to perform or that fails to perform in a timely manner under such circumstances may be terminated for default or be subject to other contractual remedies as appropriate.

CONTRACTOR VEHICLES, TRAILERS AND EQUIPMENT
All contractor owned, rented or leased vehicles, trailers and/or equipment for use on site, shall be identified with the contractor's company name, job site description, point of contact and phone number where the contractor can be reached 24 hours a day. The contractor shall ensure all referenced items are to be checked for proper operation, leaks, etc. prior to delivery to Tobyhanna Army Depot. The contractor shall coordinate with the Contracting Officer's Representative or assigned Government point of contact prior to leaving any of the referenced items unattended at the site.

INSURANCE REQUIREMENTS - CONTRACTING OFFICER'S NOTICE
The contractor shall procure and maintain during the entire period of his performance under this contract the following minimum insurance:

(a) WORKER'S COMPENSATION AND EMPLOYER'S LIABILITY. Contractors are required to comply with applicable Federal and State Worker's Compensation and Occupational Disease Statutes. If the Employer's liability section of the insurance policy, except then contract operations are so commingled with a contractor's commercial operations that it would not be practical to require this coverage. Employer's liability coverage of at least $100,000 shall be required, except in states with exclusive or monopolistic funds that do not permit Worker's Compensation to be written by private carriers.

(b) GENERAL LIABILITY. Bodily Injury Liability Insurance Coverage written on the comprehensive form of policy of at least $500,000 per occurrence shall be required. Property Damage Liability Insurance is not normally required.

(c) AUTOMOBILE LIABILITY. Automobile Liability Insurance written on the comprehensive form of policy is required. The policy shall provide for Bodily Injury and Property Damage Liability covering the operation of all automobiles used in connection with performing the contract. Policies covering automobiles operated in the United States shall provide coverage of at least $200,000 per person and $500,000 per occurrence for Bodily Injury and $20,000 per occurrence for Property Damage. The amount of liability coverage on other policies shall be commensurate with any legal requirements of the locality and sufficient to meet normal and customary claims.

16. CONTRACTOR SAFETY STANDARDS FOR TYAD
(Updated June 2024)

SERVICE or EQUIPMENT INSTALLATION CONTRACTOR SAFETY STANDARDS FOR TYAD

Below are some "installation specific" safety standards which contractors are required to adhere to while performing any work on Tobyhanna Army Depot. It is the contractor's responsibility to always incorporate these particular requirements into their daily operations while under contract and to institute these specifics into their overall site safety controls. These do not substitute or deflect in any way the requirement to institute and adhere to all OSHA 29 CFR 1910 & 1926 and USACE EM385-1 safety standards, as well as all applicable contractual Federal Acquisition Regulations while performing the work under this contract on Tobyhanna Army Depot.

It is the responsibility of the contractor to enforce the below requirements as well as all current revisions of the above referenced standards. Failure to enforce safety requirements on the depot can result in immediate stop works, safety stand down orders and even complete termination of the contract.

Important Phone Numbers:
Fire or Emergency: 911
Fire Department – (Non-Emergency): 570-615-7300
Security: 570-615-7550
Safety Office - 570-615-7027

S1.0 THE CONTRACTOR IS SOLELY RESPONSIBLE AND LIABLE FOR JOB- SITE SAFETY

1. Review of the project plans and other documents by any government appointed official does not constitute an acceptance of Federal responsibility or liability for the adequacy of the safety measures identified for the job or for the Contractor's compliance with all applicable safety rules and regulations.

2. The Contractor remains solely responsible and liable for job-site safety at all times during the term of the contract. No outside personnel (other than ET, COR, TYAD Safety, or other government appointed contract personnel) are allowed onto the job site at any time during contract without prior permissions and/or escort by the ET, COR or KO.

3. Tobyhanna Army depot has restrictions on the use of fossil fuels in all buildings. When it is determined that electric or other alternatively fueled vehicles are insufficient or unavailable for movement of material within the buildings on the installation, fossil-fueled vehicles can be used only when all alternatives of accepted electric material movement equipment have been considered, and the last option is use of a fossil fuel vehicle or project failure. This includes any situations where safety of employees and equipment may be jeopardized. Convenience will never be an acceptable justification to utilize fossil-fueled vehicles. The Government Designated Representative (GDR) will notify the appropriate supervisor(s) or leader(s) in the area where the need to utilize a fossil fuel vehicle is required. The supervisor / leader will take the necessary steps to notify the employees in the work area of the option

to leave the area during the material movement process. The GDR and Contractor will also document all actions the Contractor took to utilize electric or other alternatively fueled vehicles. The documentation will specifically state why electric or alternatively fueled vehicles are insufficient and all mitigation efforts used to minimize the effects of fossil-fueled vehicles to include information as to any appropriate scrubbers that can be utilized along with any ventilation or exhaust systems that can be used to externally vented to the outside of the facility. The documentation must be submitted to the TYAD Safety office for approval prior to any equipment being used. A final copy of the Approved documentation will be provided to the TYAD Safety Division.

4. All direct construction supervision, including subcontractors, must as a minimum, complete an OSHA 30-hour safety course for construction. These certificates must be submitted as part of the site safety plan BEFORE construction activities start.

5. **RED CARD PROGRAM** - Tobyhanna Army Depot utilizes a program to allow employees to stop an action which is considered unsafe. Employees carry small red cards that can be thrown down when employee sees any action they consider unsafe. Work ceases until TYAD management can decide if it is safe or not. Contractors are required to comply with this program. If a TYAD employee presents the red card to the contractor, the contractor must stop working until the COR and Safety Office can determine if there is a safety hazard or not.

6. All contractor personnel, to include subcontractors, vendors, suppliers, or visitors, when performing any type of work activity, or gaining access through an occupied area or shop - must have all applicable PPE that is required for the area to include, but not limited to, hardhats, safety glasses, hearing protection, and respirator protection. Notify both COR & Area Supervision, for access approval and prior to any work activities being performed.

S2.0 CODE COMPLIANCE

As noted in Tobyhanna Army Depot Safety Regulation 385-1 - TYAD follows all requirements and is certified as an ISO 45001 facility. Accordingly, the contractor shall comply with and maintain these high safety standards and provide full compliance with all applicable safety regulations and standards. The Contractor shall maintain, monitor, and enforce employee compliancewith all applicable rules, regulations, and codes.

Tobyhanna Army Depot requires full contractor compliance with all safety regulations and standards. The Contractor is responsible for maintaining, monitoring, and enforcing all rules, regulations, and codes, by ALL personnel working for the contractor including all subcontractors. These codes include:

Title 29 Code of Federal Regulations-1910 (OSHA General Industry Standards)

Title 29 Code of Federal Regulations-1926 (Construction Industry Standards)

National Fire Protection Association Codes

Uniform Facilities Code

Unified Facilities Guide Specifications (UFGS)

USACE EM385-1-1

All other codes as required to maintain safety standards.

S3.0 JOB HAZARD ANALYSIS (JHA) / ACTIVITY HAZARD ANALYSIS (AHA)

Prior to the start of work, the contractor shall prepare a JHA / AHA for each phase of work that will be done under the contract. This will be completed in accordance with all applicable OSHA safety requirements. A phase is any operations involving a certain type of work. Examples include demolition, excavation, masonry work, concrete pouring, roofing, and electrical wiring. Work done by each subcontractor is also a phase. The JHA / AHA will:

1. List the activity being performed and identify the sequence of work steps.

2. List the hazards associated with each step, and the procedures and training required to eliminate or reduce the risk to an acceptable level.

S4.0 SITE SPECIFIC SAFETY & ACCIDENT PREVENTION PLANS

Contractors will submit a written Site-Specific Safety / Contractor Accident Prevention Plan prior to beginning activities. The plan must follow the guidelines in OSHA 29 CFR 1926 or 1910 as applicable.

1. Who will implement the plan, including who is responsible for safety and accident prevention?

2. A means for coordinating and controlling subcontractors and suppliers.

3. Safety training.

4. Who will investigate accidents?

5. Emergency response.

6. Job site cleanup and safe access.

7. Public safety requirements such as signs and barricades.

S5.0 FALL PROTECTION

1. Personnel performing works at a height of six feet or more from any lower level shall comply with OSHA 29 CFR 1926.500 through 1926.503 safety standards for any type of new installations. OSHA 29 CFR 1910-28 shall apply to any type of servicing on previously installed equipment.

2. There are currently no certified anchorage points or permanent guardrail systems on any TYAD roof. Most roofs of the depot are more than 20 feet high.

3. Contractors and their employees must be protected from falling off the edge of the roof, regardless of how briefly the employees are at the edge of the roof.

S6.0 ELECTRICAL SAFETY

Follow the current Tobyhanna Army Depot standard. This standard is based on NFPA 70E.

Copies of the TYAD Electrical standard may be obtained by request from the ET, COR or KO on the project or service contract.

S7.0 DEPOT REQUIRED PERMITS / APPROVALS

Permits are required for:

1. **Excavation**: Contractors must utilize the PA 1 Call system and also coordinate with contracting officer representative (COR) for a TYAD dig permit for any excavations prior to any groundbreaking including digging, drilling and stake driving.

2. **Trenching**: Contractors will adhere to OSHA requirements and receive approval from TYAD Safety office before any trenching deeper than 5 feet is performed. Notifications of such activities will be provided to the COR along with required plans, requests and permits prior to any trenching works being performed. Contractor is responsible for field verifying all utilities and underground entities. Excavation or trench greater than 18 inches in depth that remains open greater than (1) day OR left unattended for any period of time shall have a safety orange construction fence installed around the perimeter of the opening. Caution tape or cones for this purpose is unacceptable.

3. **Confined Space**: All confined space works on TYAD are to be considered as permit required entries, (unless reclassified by the TYAD Safety office,) and Contractor must first request and be provided with an approved "signed" permit from TYAD Safety Office prior to any confined space entry. Contractor will refer to all OSHA requirements and must have a confined space plan and all appropriate safety equipment. Contractor is to request approval from TYAD Safety office through COR prior to entry.

4. **Burning / fire***:* Hot work permits must be obtained daily from the Fire Department.

5. **Cranes:** For use of cranes and all WHE and LHE at TYAD see S.10.13

6. **Army Radiation Permits:** Non-Army agencies (including other military Services, vendors, and civilian contractors) require an Army Radiation Permit (ARP) to use, store, or possess ionizing radiation sources on TYAD (see 32 CFR 655). Non-Army applicants will apply by letter with supporting documentation to the Depot Commander. The letter should be submitted so that the Depot Commander receives the application at least 30 days before the requested start date of the permit. The ARP application will specify start and stop dates for the ARP and describe the intended use of the radioactive material.

For sealed sources of radiation, i.e. Moisture Soil Density Gauges, the application must include a valid Nuclear Regulatory Commission (NRC) Radioactive Materials License that allows the applicant to use the source as specified in the ARP application, current leak test certificates, operator training records and calibration certificates for any equipment that may be utilized at

TYAD.

For machine produced ionizing radiation sources, i.e. X-ray Equipment, the application must include the appropriate state authorization that allows the applicant to use the source as specified in the ARP application along with operator training records and calibration certificates for any equipment that may be utilized at TYAD.

ARPs will not be issued for more than 12 months at a time.

S8.0 CONFINED SPACES

1. All Confined Spaces at TYAD are considered as Permit-Required Confined Spaces unless reclassified by TYAD Safety office after a thorough review of the area and the work being planned.

2. Contractors must first notify the COR and receive TYAD Safety office approvals prior to gaining access or opening any confined space area.

3. Contractors must fully comply with all OSHA regulations for Confined Space Operations, including on-site use and availability of a Confined Space Entry Permit / Checklists, along with prior approvals of their Accident Prevention Plan and associated Activity Hazard Analysis (AHAs) for the work to be performed.

4. The contractor and/or the COR will then notify the TYAD Safety office and TYAD Fire and Emergency Services of the Confined Space Entry timelines and schedules.

5 Upon completion, the contractor will notify the TYAD Safety Office and discuss whether any new hazards were added to the Confined Space and will submit the completed Confined Space Entry Permit / Checklist for record.

S9.0 LOCKOUT/TAGOUT PROCEDURE

1. The Contractor shall perform this work in accordance with 29 CFR § 910.147-The Control of Hazardous Energy requirements.

2. The Contractor shall notify the COR that a lockout/tag out system is going to be used.

3. The contractor is to request the COR to notify them if TYAD is using any lockouts / tag outs in the construction area.

4. The Contractor shall locate and identify all isolating devices.

5. The Contractor shall shut down the equipment normally.

6. The Contractor shall isolate all energy using appropriate methodology.

7. After all work is complete, the Contractor shall remove the energy isolating device and restore the equipment to service.

8. The Contractor shall notify the COR that energy is again restored.

S10.0 PROTECTION TO THE DRINKING WATER SYSTEM

1. The contractor will not perform any work on the depot potable water system until prior approval is obtained from the Environmental Branch and the certified operator through the Directorate of Installation Services COR.

S11.0 PROTECTION OF PERSONNEL

At no time is any depot employee to be put at risk to illness or injury. No construction actions are exempt from this requirement.

1. Where pedestrian and driver safety is endangered, use appropriate traffic barricades. Use anchor barricades to prevent displacement by wind. Institute PA DOT regulations and OSHA requirements. Notify the COR prior to beginning such work.

2. Where demolition is required, a Demolition Plan describing the steps and individual Definable Features of Works (DFOWs) as well as safety protocols, controls and oversight is required to be submitted to the COR for review prior to works being performed. Continuously evaluate the condition of the structure being demolished and take action to protect personnel working around the project site. No structural element will be left standing without sufficient support to prevent collapse.

3. Contractor shall provide personal protective equipment (PPE) and provide training for use, for all visitors requiring entry into the regulated area.

4. Subcontractors, vendors, and delivery personnel are the responsibility of the prime contract and must follow the same regulations the prime contractor does.

S12.0 FIRE AND EMERGENCY SERVICES

Hot Work Permits will be issued by Fire Department upon request. All requests must be made at least thirty (30) minutes prior to the expected start time by contracting the fire department.

1. Contractors are required to have fully charged and operable contractor provided fire extinguisher, in quantities and properly spaced per OSHA requirements that are appropriate for the type of possible fire. Fire extinguisher shall be immediately available for inspection by the fire department upon an inspection of the worksite and shall remain at the work site at all times. At a minimum, 10 pound class 4A, 60B or C multi-purpose dry chemical fire extinguisher or equivalent shall be provided.

2. A fire watch must be in place during the hot work and for 30 minutes after the HWP operation. Upon completion of the thirty (30) minute, post hot work fire watch, the contractor shall contact the fire department to report the work is complete.

3. Use of a plasma torch will require special approval by the fire department.

4. Any fire, even one that is extinguished, must be reported to the fire department.

5. Ambulance services are provided at a cost to contractors. The contractedambulance service provider will transport to the nearest hospital. Advanced Life Support, helicopter transport, etc. is all at the contractor's expense.

6. The fire department will respond to all fire, ambulance, hazmat and confinedspace incidents.

S13.0 IF AN INCIDENT HAPPENS

1. **Any serious injury or medical emergency**:
   - Immediately call emergency responders
   - Dial 911 and give accurate location and nature of emergency. Make sure to tell the operator that you are calling from Tobyhanna Army Depot.
   - Send someone to the nearest exit to assist responders.
   - Treat victims to the best of your ability.
   - Notify COR / call work order desk at 570-615-7805 if there is any blood or body fluids that must be cleaned up.

2. **In the event of the smell of natural gas:**
   - Call dial 911and give accurate location and nature of emergency. Make sure to tell the operator that you are calling from Tobyhanna Army Depot.
   - Have all personnel evacuate the area if the smell is significant.
   - Upon arrival of fire personnel, provide information as to specific equipment and exact location of potential leak.

3. **In the case of a fire**:
   - If the fire is small, use a fire extinguisher and call 911.
   - Otherwise, dial 911 and pull the fire alarm.
   - Evacuate the building.

4. **Accident/Injury Reports and OSHA 301 Reports**:

The contractor shall report to the COR and KO, - Immediately but no later than eight (8) hours, any Fatality, In-patient hospitalization, Amputation, Eye loss or Property damage over $600,000. Similarly, contractor shall report to the COR and KO, – Immediately, but no later than twenty-four (24) hours, all other accidents or near misses.

Notifications shall be followed up within five (5) working days of all incidents (reportable or not) with an OSHA 301 report including all available facts relating to each instance of personal injuries and of loss / damage to Government property.

For any accident that results in illness or injury that requires more than first aid treatment, TYAD Safety & Health Office has the option to have the contractor immediately secure the accident area until released by the accident investigation authority as designated by the Commander. If theGovernment elects to conduct an investigation of the accident, the contractor shall cooperate fully and assist authorities until the investigation is complete. If requested after the investigation is completed, the contractor shall provide a Corrective Action Plan related to the incident to the COR & KO.

COR to provide copies of all OSHA 301 incident reports and any Correction Action Plans to the

TYAD Safety office when available.

S14.0 CRANE LIFT OPERATIONS AT TOBYHANNA ARMY DEPOT

A Lift Plan for Crane Operations will be submitted to the ET/COR for approval by the TYAD Safety Office *at least 14 days* prior to a lift operation. It will conform to all applicable OSHA regulations and requirements and shall include:

1. A description of the operation to be performed, specific personnel assignments (Lift Director, Safety Coordinator, Operator, rigger, etc.) and signaling and any other details pertaining to activities during the lift.

2. A drawn Site Plan with specifications including work zone, support equipment locations, evacuation area, traffic control, electrical power line clearances (if applicable), lift path, surface conditions at crane location, etc.

3. Training/Certifications (scans) for crane operator and other lift-teammembers included in the lift operation.

4. Equipment nomenclature and a certificate of comprehensive annualinspection records, including any deficiencies and corrective action.

5. Load chart for crane, with target capacities specific to the crane/boomconfiguration and lift details clearly marked.

6. Lift calculations, including rigging components and configuration.

7. Weather conditions are always part of the ongoing safety assessment and may result in rescheduling of the lift.

8. Only approved operators are allowed to operate or move the crane for any reason.

A Production Lift Plan pertains to repetitive lifts within a work project and may be outlined with one lift plan depicting the nomenclature and general capacities of the equipment to be utilized. Besides the eight items above, the plan will also include how often the crane will be used and how traffic and personnel safety will be handled. A separate traffic plan will be required for vetting and approval prior to any lifts taking place.

S15.0 APPLICATION OF ANY ODOR PRODUCING MATERIAL

1. Any application of any paint, epoxy, adhesive or anything else that produces an offensive odor must be coordinated with the COR or POC before starting. This is particularly true of the construction work in a building occupied by Tobyhanna employees.

2. It is highly recommended that this type of work be planned for off hours, weekends, or holidays to avoid shutdowns of operations due to these offensive odors. Odors during curing times for products shall also be considered.

S16.0 CONTROL OF RESPIRABLE CRYSTALLINE SILICA

1. Contractors are required to follow the OSHA standard 29 CFR 1926.1153 which requires employers to limit worker exposures to respirable crystalline silica and to take steps to protect their workers and TYAD personnel.

2. Contractors shall establish and implement a written exposure control plan that identifies tasks that involve exposure and methods used to protect workers, including procedures to restrict access to work areas where high exposures may occur.

3. Restrict housekeeping practices that expose workers to silica by utilizing engineering controls such as local exhaust ventilation, using "wet spray" methods when cutting or drilling and providing proper respiratory protection and other PPE as applicable.

S17.0 OPERATION & MAINTENANCE MANUALS

1. The Contractor shall provide to the COR a copy of all Operation and Maintenance (O&M) Manuals and Parts Manuals for any equipment provided.

2. The O&M Manuals shall contain a complete set of mechanical drawings and diagrams; a complete set of electrical drawings and diagrams including electrical schematics; logic circuitry descriptions; spare parts list; operator and maintenance instructions for operating, testing, and maintaining all system components.

3. The O&M Manuals shall contain complete detailed instructions and schedules for system PM.

4. The O&M Manuals shall contain a detailed description of the lock-out/tag-out procedures needed to bring the system and individual components to a zero-energy state for maintenance and servicing.

5. The O&M data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

IN ANY CONFLICT BETWEEN STANDARDS, THE STRICTER OF THE TWO SHALL APPLY. IF UNSURE CONTACT THE COR / ET VIA PROPER RFI PROTOCOLS FOR VERIFICATION THROUGH THE TYAD SAFETY OFFICE.

17. EXCAVATION REQUIREMENTS

Tobyhanna Army Depot Standard Operating ProcedureFor Excavation/Excavation Permits

This document is intended for use at Tobyhanna Army DepotContractors are expected to comply with Pa state law. (Pennsylvania One Call System)

This procedure applies to any excavation deeper than 6 inchesincluding boring, drilling, augering and driving stakes.

1. The excavator is required to outline the proposed excavation site with whitemarkings before requesting a permit
2. Call Pennsylvania One Call System at 800-242-1776 or 811
3. Pa One Call will notify Tobyhanna (Facility Owner) via fax.
4. TYAD will mark utilities and provide a signed permit to the excavatorif approved.
5. TYAD has a maximum of ten working days to approve each permit
6. Any disapproval stops the permit process and requires a resubmission
7. Excavator must have a signed permit on the job site at all times
8. The permit is valid for thirty (30) days
9. The excavator is responsible for maintaining any markings made by thefacility owner.
10. The excavator is expected to hand dig within thirty (30) inches of markedfacilities. (See figure on page 2).

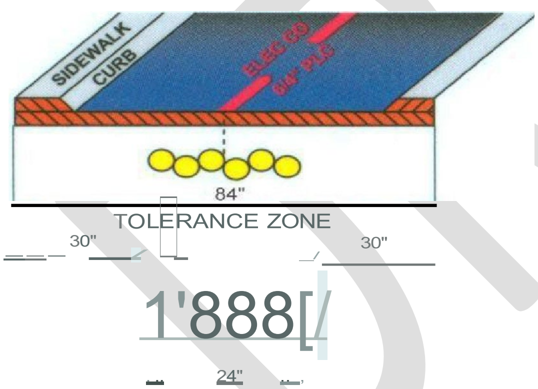
Temporary Facility Marking Responsibility by Facility Owners is:
To mark, stake, locate or otherwise provide the position of the underground utilities at the site within 30 inches* horizontally from the outside wall of the utilities in such a manner so as to enable the excavator, where appropriate withinthe tolerance zone, to employ prudent techniques, such as hand digging, to determine the precise location of the utility lines.

*Note: The 30 inch horizontal requirement is specific to Tobyhanna ArmyDepot

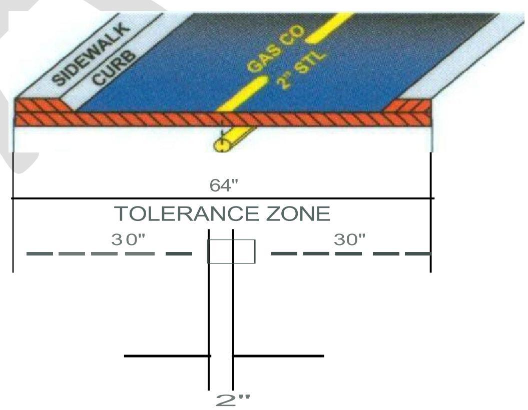
APWA/CGA TEMPORARY MARKING COLOR CODE (ANSI Standard 2535.1)

WHITE	Proposed Excavation
PINK	Temporary Survey Markings
RED	Electric Power Lines, Cables, Conduit and Lighting Cables
YELLOW	Gas, Oil, Steam, Petroleum or Gaseous Materials
ORANGE	Communication, Alarm or Signal Lines, Cables or Conduit and Traffic Loops
BLUE	Potable Water
PURPLE	Reclaimed Water, Irrigation and Slurry Lines
GREEN	Sewer and Drain Lines

TOLERANCE ZONE



LARGE PIPE OR MULTIPLE DUCTS



SMALL PIPE OR CABLES

THESE REQUIREMENTS ARE SPECIFIC TO TOBYHANNA ARMY DEPOT

PROCEDURES FOR EXCAVATIONIN

ACCORDANCE

WITH PENNSYLVANIA ONE CALL (PA One Call) REQUIREMENTS

Tobyhanna Army Depot

TOBYHANNA ARMY DEPOT IN-HOUSE EXCAVATION PERMIT

For Excavation And Other Workin the

Vicinity of UtilitiesCALL 911 FOR

EMERGENCIES

CHECK APPROPRIATE BOXES, FILL IN ALL SPACES AND SUBMIT TO THE ENGINEERING BRANCH POC AT LEAST WORKING 3 DAYS PRIOR TO START OFEXCAVATION. EXPLAIN ANY "NO" RESPONSE TO ANY ITEM BELOW IN SECTION 6.

PROJECT:__________________JOB ORDER NUMBER:________________________

PROPOSED EXCAVATION START DATE/COMPLETION DATES:

____

1. EXCAVATION REQUIREMENTS DEFINED. - ☐ YES ☐ NO

2. PROPOSED EXCAVATION AREA MARKED WITH WHITE LINES.

_☐ **YES** ☐ **NO**

3. CONTACT PA ONE-CALL SERVICE: 1-800-242-1776. - ☐ **YES** ☐ **NO:**

Date & Time____________________________________Serial No.

_(PA ONE-CALL covers all Natural Gas lines)

4. CONTROLCONSIDERATIONS

a. SAFETY MEASURES ARE IN PLACE TO ASSURE EXCAVATION AND PROCEDURESMEET OR EXCEED ALL OSHA AND OTHER REQUIREMENTS.

\- ☐ YES ☐ NO

b. LOCATION OF SHUT-OFF OR ISOLATION VALVES HAVE BEEN ESTABLISHED.

\- ☐ YES ☐ NO

c. SOIL EROSION AND SEDIMENTATION CONTROLS WILL BE IMPLEMENTED

\- ☐ YES ☐ NO

5. NOTES:

a. Work shall always proceed in an orderly and effective manner topreclude potential damage or problems.

b. When utilities have been located in the area to be excavated,hand digging will be used within 30 inches until the exact locations of all lines have been determined.

c. If differing site conditions are determined to exist, contactthe Directorate of Installation Services Engineering Branch POC for possible additional precautionary measures to be employed.

6. REMARKS:

____

____

____

____

____

____

7. POC FROM Roads & Grounds Branch (equipment operator orperson knowledgeable about the excavation):

____

8. ____ ____________________

AUTHORIZED SIGNATURE (BRANCH CHIEF/LEADER) DATE

PROCEDURES FOR EXCAVATION

IN ACCORDANCE

WITH PENNSYLVANIA ONE CALL (PA One Call) REQUIREMENTS Tobyhanna Army Depot

THE FOLLOWING SECTIONS TO BE COMPLETED BY ENG BRANCH POC AFTER RECEIPT OF PERMIT REQUEST:

9. DATE RECEIVED ________________________________

10. DATE UTILITIES MARKED ___________________________________

11. UTILITIES IDENTIFIED ON-SITE:

☐ NONE

☐ ELECTRIC ☐ CATV

☐ NATURAL GAS ☐ SEWER

☐ WATER (active)

☐ TELECOMM

☐ STEAM (abandoned)

POTENTIAL ASBESTOS

☐ OTHER ____________________

12. CONTROLSREQUIRED:

☐ NONE

☐ HAND EXCAVATE TO LOCATE UTILITY

☐ OTHER SPECIAL CONDITIONS OR PRECAUTIONARY MEASURES TO BE IMPLEMENTED:

____

____

PERMIT APPROVED - ☐ YES ☐ NO

AUTHORIZED SIGNATURE (Chief Eng Branch or Designated Alternate)
____DATE____________________

EXPIRATIONDATE:________________________________

PROCEDURES FOR EXCAVATION
IN ACCORDANCE
WITH PENNSYLVANIA ONE CALL (PA One Call) REQUIREMENTS
Tobyhanna Army Depot

TOBYHANNA ARMY DEPOT EXCAVATION
PERMIT & RISK ASSESSMENT

For Excavation and Other Work in the Vicinity of Utilities

EMERGENCY PHONE NUMBER - 615-7300

CHECK APPROPRIATE BOXES, FILL IN ALL SPACES AND SUBMIT TO THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) AT LEAST 3 WORKING DAYSPRIOR TO START OF EXCAVATION. WORK SHALL NOT PROCEED UNTIL PERMITIS ISSUED BY DIRECTORATE OF INSTALLATION SERVICES (D/IS). COPY OFPERMIT SHALL BE KEPT ON SITE AT ALL TIMES. EXPLAIN ANY "NO" RESPONSE TO ANY ITEM BELOW IN SECTION 12.

PROJECT:_____________**CONTRACT NUMBER:**_______________________________

PROPOSED EXCAVATION START DATE/COMPLETION DATES:

____

1. EXCAVATION DETAILS AND DRAWINGS ESTABLISHED. - ☐ YES ☐ NO

2. PROPOSED EXCAVATION AREA MARKED WITH WHITE LINES.- ☐ YES ☐ NO

3. CONTACT PA ONE-CALL SERVICE: 1-800-242-1776. -Date & Time___ ☐ YES ☐ NO:

Serial No. ____

(PA ONE-CALL also covers all GAS lines on the installation)

4. UTILITIES IDENTIFIED ON-SITE:

☐ NONE ☐ ELECTRIC ☐ CATV ☐ GAS

☐ SEWER ☐ WATER ☐ TELECOMM ☐ STEAM (abandoned)

☐ OTHER __________________

5. DATE UTILITIES MARKED AND METHOD OF MARKING

PA ONE CALL- ____

OTHER - ____

6. LEVEL OF RISK: (Based upon personnel safety and consequences ofutility outages.)

☐ SEVERE: Excavation required within 3 feet of GAS line.

☐ MODERATE: Excavation required within 2 feet of another utility.

☐ MINIMAL: No excavation required within 2 feet of any utility.

7. CONTROLSREQUIRED:

☐ NONE

☐ HAND EXCAVATE TO LOCATE UTILITY

☐ OTHER____

8. CONTRACTOR'S EMERGENCY NOTIFICATION PHONE NUMBER OR PAGER:

____

9. CONTRACTOR'S CONTROL CONSIDERATIONS

a. SAFETY MEASURES ARE IN PLACE TO ASSURE EXCAVATION AND PROCEDURESMEET OR EXCEED ALL OSHA AND OTHER REQUIREMENTS.

- ☐ YES ☐ NO

b. LOCATION OF SHUT-OFF OR ISOLATION VALVES HAVE BEEN ESTABLISHED.

- ☐ YES ☐ NO

c. SOIL EROSION AND SEDIMENTATION CONTROLS WILL BE IMPLEMENTED

- ☐ YES ☐ NO

d. CONTINGENCY PLANS AND ACTIONS TO BE TAKEN IN THE EVENT OF ANINCIDENT RESULTING IN UTILITIES INTERUPTION OR DAMAGE TO FACILITIES.

- ☐ YES ☐ NO

(Attach additional sheets as necessary).

10. NOTES:

a. Receipt of a permit from the government does not relieve the contractor from responsibility for damage to utilities. Work shallalways proceed in an orderly and effective manner to preclude potential damage or problems. The contractor will be liable for alldamages to persons or property that occurs as a result of the contractor's fault or negligence.

b. When utilities have been located in the area to be excavated,hand digging will be used within 30 inches until the exact locations of all lines have been determined.

c. The Government reserves the right to specify additional precautionary measures to be employed if differing site conditionsare determined to exist.

CONCUR WITH NOTES: ☐ YES ☐ NO:

11. Remarks:

THE INFORMATION NOTED ABOVE IS ACCURATE AND TRUE AND THE WORK ISREADY TO PROCEED:

AUTHORIZED CONTRACTOR SIGNATURE DATE PROCEDURES FOR EXCAVATION INACCORDANCE WITH PENNSYLVANIA ONE CALL (PA One Call) REQUIREMENTS

Tobyhanna Army Depot

****

**************TO BECOMPLETEDBY TYAD REPRESENTATIVE*********

PERMIT APPROVED - ☐ **YES** ☐ **NO** (see cover letter for reasons and corrective actions required with time frames)

AUTHORIZED D/IS SIGNATURE AND DATE:___________________________________EXPIRATION
CONDITIONS OR PRECAUTIONARY MEASURES TO BE IMPLEMENTED:

____

____

____

____

____

____

THE FOLLOWING SECTIONS TO BE COMPLETED BY ENG DIV POC AFTER RECEIPT OF PERMIT REQUEST:

12. DATE RECEIVED _________________________________

13. DATE UTILITIES MARKED ___________________________________

14. UTILITIES IDENTIFIED ON-SITE:

☐ NONE

☐ ELECTRIC ☐ CATV

☐ NATURAL GAS ☐ SEWER

☐ SEWER ☐ TELECOMM

☐ STEAM (abandoned)

POTENTIAL ASBESTOS

☐ OTHER ____________________

15. CONTROLSREQUIRED:

☐ NONE

☐ HAND EXCAVATE TO LOCATE UTILITY

☐ OTHER SPECIAL CONDITIONS OR PRECAUTIONARY MEASURES TO BE IMPLEMENTED:

____

____

____

____

____

PERMIT APPROVED -
AUTHORIZED SIGNATURE (Chief Eng Branch or Designated Alternate)
____DATE______________________
EXPIRATIONDATE: ____

NOTICE to ALL CONTRACTORS INTALLING NEW UNDERGROUND UTILITIES

All new underground utilities within the construction area shall be located and identified onas-built drawings using the Pennsylvania (North) State Plane coordinate system (North American Datum1983 Feet (NAD83)) to obtain a NORTHING and EASTING.While vertical surveys will use the North American Vertical Datum (NAVD) 1988 coordinate system for anELEVATION. Refer to Scope of work Section 17 Deliverables Section C paragraphs 1 through 9 fordetails.

Providing survey date is not a substitute for othermarking methods. Contractors are responsible for implementing other markings such as tracing wire ortape as required

18. ENVIRONMENTAL REQUIREMENTS

Standard Specifications for Projects Under the National Environmental Policy Act in Accordance with 32 CFR 651 Proponent ELTY-ISP-N (9 Jun 2025)

Spill Response Procedures 1-2-3

1. Stop work
2. Call 911
3. Evacuate Area

*********Damaged Asbestos?*********
Environmental Questions?
Call Environmental Branch (EB) at
570-615-7098

1. Air Pollution Control

The contractor must control fugitive emissions, including dust, during the course of their contract. The contractor must obtain approval from the Environmental Branch (EB) prior to exhausting equipment to the outside. The contractor must not allow any pollutant or particulate matter to be released to the atmosphere at levels that are visible from outside of Tobyhanna Army Depot (TYAD). The contractor must not perform work that will release pollutants or particulate matter to the atmosphere when the wind speed exceeds ten miles per hour and will result in adverse effects to the surrounding areas. TYAD will monitor the wind speed; it is the contractor's responsibility to obtain wind speed information through the Contracting Officer Representative (COR).

2. Asbestos

The contractor must ensure that all materials used in the performance of this contract are asbestos-free.

Unless specified in the contract, the contractor must not disturb any existing Asbestos-Containing Material (ACM) in the performance of this contract. If ACM, or suspect ACM might be disturbed in performance of this contract, the contractor must avoid encountering the material and immediately notify the COR and the Contract Administrator in writing. The COR must coordinate with EB to have the material tested to determine if there is ACM. If the material is determined to be ACM and the contractor cannot avoid disturbing the material, the COR will notify the Contracting Officer. The Contracting Officer will direct a change pursuant to the contract clauses entitled "Changes" and "Differing Site Conditions." If ACM, or suspect ACM has been disturbed, the contractor must immediately notify the COR and call the EB. The contractor will shut down and not move any equipment or supplies near the damaged ACM. The contractor will evacuate all non-contaminated contractor personnel from the immediate vicinity. Any contractor personnel thought to be contaminated with asbestos must remain in the area until the EB responds. If the damaged material is determined to be ACM and there is potential for further damage, the Contracting Officer will direct a change pursuant to the clauses of the contract clauses entitled “Changes” and “Differing Site Conditions.”

Asbestos abatement required under the contract as originally awarded must be in accordance with United Facilities Guide Specification 02 83 14 00 10 and as otherwise required in the contract. Asbestos abatement not required under the contract as originally awarded, can be incorporated into the existing contract via contract modification, or by the Government taking responsibility for the asbestos abatement. The method of acquiring the abatement is at the discretion of the Government. The contractor must have an asbestos abatement work plan that has been approved by EB prior to beginning any asbestos abatement work.

3. Backflow

The contractor shall have a backflow prevention device installed on all contractor equipment that is connected to TYAD’s water distribution system. The contractor shall have a water meter installed to monitor water consumption during all phases of the contract. Water use will be reported monthly to the COR.

4. Burning

The contractor must not burn refuse and debris anywhere on TYAD.

5. Buy Recycled-Content Materials

The Contractor must comply with Resource Conservation and Recovery Act (RCRA) Section 6002 (42 U.S.C. 6962, Federal Procurement) in the acquisition of materials with recycled content. Specific designated items in this contract for which recycled content standards have been established have been set forth in the specification (e.g., insulation, roofing materials, carpet, carpet pad, paint, floor tiles, shower and restroom dividers). Recovered Material Certification: As required by the RCRA, the contractor must certify that the percentage of recovered materials to be used in the performance of the contract will be at least the amount required by the applicable contract specifications (see Federal Acquisition Regulation (FAR) Provision 52.223-4, Recovered Material Certification). Prior to application for final payment, the contractor must provide a report in accordance with FAR Clause 52.223-9, Estimate of Percentage of Recovered Material Content for Environmental Protection Agency (EPA) Designated Items, to the Contracting Office. Compliance with this program does not relieve the contractor from meeting all other specification requirements.

6. Cultural Resources

The contractor must not adversely affect any property listed on the National Register of Historic Places (NRHP) or properties eligible for inclusion on the NRHP without consultation and approval from the EB through the COR. If there is a discovery of any historic properties, including archeological sites and graveyards, work will cease immediately until requirements of National Historic Preservation Act, as amended, have been met. All archaeological artifacts found at TYAD or TYAD-controlled properties are U.S. Government property until a determination is made otherwise.

7. Demolition Notification

If a project involves the demolition of any load-bearing structural members, whether or not asbestos is present, 25 days prior to the demolition, the Commonwealth of Pennsylvania Asbestos Abatement and Demolition/Renovation Notification Form 2700-FM-BAQ0021 must be submitted to EB as specified in the Unified Facilities Guide Specification 02 82 13.00 10.

8. Drinking Water

The contractor must not perform any work on the TYAD potable water system prior to obtaining approval from the EB and the Installation Planning and Maintenance Division certified operator through the COR. If a permit is required due to construction or proposed chemical feed changes, it will be the responsibility of the contractor to obtain all permits associated with the project. Permit applications will first be reviewed and approved by the Environmental Branch before being submitted to regulatory authorities. Any piping or additions added to the TYAD water system must be disinfected following "American Water Works Association circular C651-14 "Disinfecting Water Mains." Construction will be kept outside of drinking water wellhead protection zones when feasible. All new buildings and remodeled buildings will be equipped with a water meter that can easily be read from the exterior of the building.

9. Endangered Species

The contractor is responsible for meeting requirements of the Endangered Species Act of 1973. The contractor must not disturb any endangered species, their habitat or offspring during the implementation of this contract.

10. Emergency/Spills

All emergencies and spills must be reported to the TYAD Fire Department by calling 911 from a TYAD phone or (570) 615-7300. If a 911 call is placed on a cell phone, the call will go to the Monroe County Emergency Office. Notify the Monroe County Emergency Office that you are at TYAD, and the call will be forwarded to TYADs Fire Department. The contractor must ensure all personnel working on site are trained in the proper procedure according to 29 CFR 1910.120 (if applicable) to initiate a spill response to handle the hazardous substances they are working with. The contractor will take the necessary actions to prevent and contain spills of hazardous materials.

11. Energy Resilience and Mission Readiness

The Energy Policy Act of 2005 section 109 and the Energy Independence and Security Act of 2007 require all new construction at federal facilities to be 30% better than ASHRAE 90.1. Energy Policy Act of 2005 section 104 and the Energy Independence and Security Act of 2007 require all new equipment to be Energy Star qualified when available. This is applicable to heating, ventilation and air conditioning (HVAC) equipment, plumbing, building materials, lighting, commercial food service equipment and appliances.

12. Environmental Automation and Control Systems
The contractor will not alter, modify, remove or tamper with any environmental automation or control system unless previous arrangement has been made with EB. This includes sensors, programmable logic controllers, equipment housing, power supplies, meters or other hardware/sensor technology.

13. Erosion and Sedimentation Control
The contractor shall protect streams, lakes and wetlands from sediment discharges caused by the contractor's activities. The contractor shall also provide (where applicable) an erosion and sedimentation control plan in compliance with state and local laws and regulations, to the COR for approval prior to executing any soil-disturbing activities. The contractor must submit a National Pollutant Discharge Elimination System Permit (NPDES) to the Pennsylvania Department of Environmental Protection (PADEP) prior to any construction activity that encompasses more than one acre. All permits must be reviewed and approved by the EB prior to being sent to any regulatory authority. The contractor shall not disturb any wetlands. The contractor is to place silt sock around dirt, rock, or construction debris to ensure runoff does not occur. The contractor will place silt sock, or drain covers around/in storm drains surrounding the construction and downstream of the site. Contractor will remove all silt fencing and other temporary control measures once the site is stabilized. Contractor will remove all spoils from TYAD unless a previously approved disposal site has been established. The contract COR will be responsible for management and control of the spoil's disposal site.

14. Fluorescent and Mercury-Bearing Lamps
The contractor is to collect, containerize, manage and recycle fluorescent and mercury-bearing lamps in accordance with 40 CFR 273, Standards for Universal Waste Management. A copy of the manifest/Bill of Lading must be given to the EB through the COR five days in advance of shipment by the contractor so that it can be reviewed for accuracy and completeness. The contractor must install low mercury bulbs when available.

15. Hazardous Materials Stored and Labeled
The contractor must ensure all hazardous materials (HM) at the work site are properly stored and labeled. The contractor must not leave any HM behind at the completion of the job for any reason. HM must not be stored outside without adequate secondary containment and shelter.

16. Hazardous Waste
The contractor must ensure that all hazardous wastes (HW) at the work site are properly stored and labeled in a pre-approved location designated by the EB. The contractor must provide copies of any shipping documents for HW/Universal Waste/Toxic Substances Control Act waste. If the TYAD EPA number is being used for shipping purposes, only EB is authorized to sign a manifest, and the manifest (or copy) must be supplied prior to the day of shipment for review. The EB will keep original

copies. If HW or waste requiring special handling (e.g., asbestos) is being turned over to the government for disposal, the contactor must notify the EB through the COR when the waste is ready to be moved.

17. ISO 14001

TYAD is an ISO 14001 certified facility. All contractor and subcontractor employees on site must comply with TYAD Regulation 200-5, "Environmental Management System (EMS)." The contractor may obtain these regulations through the project COR.

18. Limit of Disturbance

The contractor must confine the limit of disturbance of the project to the smallest area possible.

19. Mercury-Bearing Equipment

The contractor must not install any equipment, switches, or devices (including thermostats) that contain mercury or lead.

20. Safety Data Sheets (SDSs)

The contractor must submit SDSs for all hazardous materials proposed for use, including paints, solvents, adhesives, etc., to the EB through the COR five working days prior to material being brought on post. As of June 1, 2015, all Material Safety Data Sheets (MSDS) must be replaced with new Safety Data Sheets (SDS). EB will no longer be accepting older legacy MSDS sheets, and the contractor must submit the current SDS for the hazardous material. The contractor must keep a copy of all SDSs required for the project at the jobsite.

21. Migratory Bird Protection

The contractor is responsible for meeting requirements of the Migratory Bird Treaty Act of 1918 (as amended). The contractor must not disturb any migratory bird, their nesting area or offspring during the implementation of this contract.

22. National Pollutant Discharge Elimination System (NPDES) Permits

The contractor must not perform any work on existing NPDES structures or treatment units unless previously approved in writing by the Installation Planning and Maintenance Division certified operators and the Environmental Branch. This includes work within the sewage treatment plant, sewage lift stations, sewage conveyance pipes, Industrial Operations Facility pretreatment plant, and storm water sewer systems. If a permit is required, it will be the responsibility of the contractor to obtain all permits prior to work being performed. All permit application packages must first be reviewed by the Environmental Branch prior to being sent to any regulatory authority. All liquids (chemical, cleaner, and lubricants) in containers, drums, and IBC totes are to be stored on secondary containment at all times. The container shall be labeled with the contents inside and an SDS sheet must be provided to EB.

23. Net Zero Water
The contractor will minimize the use of potable water during the construction project. Water used during the construction project will be monitored and measured using portable water meters if possible. Installed restroom equipment will consist of high-efficiency fixtures that use reduced volumes of water. Use of flow regulation devices must be approved by EB prior to installation. Employ strategies that in aggregate use 20% less water than the water use baseline calculated for the building after meeting the Energy Policy Act of 1992 fixture performance requirements. Automatic hands-free flushometers will not be used. Water-free urinals will not be used. Flushometers will be the piston variety that fail in the closed position. New building construction and renovations will include installation of a water meter that is capable of being read from the outside of the building.

24. Noise
The contractor must not allow the noise level to exceed 65 decibels at any point outside TYAD property. If noise levels exceed 65 decibels, a plan must be prepared by the contractor to mitigate the noise levels and submit to the EB for approval through the COR. The contractor will monitor the fence line to confirm this limit.

25. Ozone Depleting Substances (ODS)
The contractor must be responsible for ensuring that all personnel who perform maintenance and repair activities on refrigeration equipment have been trained and certified by an EPA-approved Section 608 program. The contractor must not use Class I or Class II ODS or install equipment that contains Class I or II ODS. Beginning 1 Jan 26, the contractor must adhere to all AIM Act regulations regarding Hydrofluorocarbon refrigerants, Title 40, Chapter I, Subchapter C, Part 84.

25. Paints
The contractor must not use paints containing zinc chromate or strontium chromate pigments and paints must not contain a concentration of lead greater than 0.009 percent. Coatings used within permitted paint spray booths must not have a Volatile Organic Compound (VOC) content greater than 3.50 pounds per gallon.

26. Pest Management
At no time during the execution of this contract must the contractor provide a food source or harborage for any pests. The contractor must coordinate through the COR to the EB prior to the application of any pesticide (PA state license is required for pesticide applications). All pesticides are required to be approved by the EB. The contractor must report all usage of pesticides the day they are applied through the COR to the EB. After completion of the contract, the contractor must ensure there is no passage for pests to enter facilities or structures related to work performed by the contractor.

27. Polychlorinated Biphenyls (PCBs)
The contractor must not bring items containing PCBs onto TYAD. Light ballasts that are clearly marked "Contains no PCBs," or that are marked with a manufacture date after 1978 must be

disposed of by the contractor as construction demolition debris. Any light ballast that is not marked as containing no PCBs that has a manufacture date prior to 1978, or that cannot be determined whether it contains PCBs, must be disposed of by the contractor at an approved and licensed facility for PCBs. The contractor must submit a shipping manifest and certificate of disposal of the PCB-containing items to the EB through the COR.

28. Recycling
The contractor must comply with TYADs general recycling plan for recyclable materials such as aluminum, steel, cardboard, paper, plastic and wood. The contractor should contact the EB for additional information on the recycling of materials through the COR. The COR will coordinate with EB to have the contractor recycle metals, cardboard, etc., through TYADs recycling program. All Construction and Demolition (C&D) material transferred from a construction project into the TYAD recycling program must be segregated and material type and weights submitted to the project COR for consolidation. Any material entering the TYAD recycling program that is not documented and properly reported will be absorbed into the TYAD recycling program and not count towards the C&D requirement of the project. Clean wood shall be recycled by the contractor off TYAD property.

29. Refuse and Construction Demolition Debris Removal and Disposal
During the performance of all construction, renovation and demolition projects, a minimum of 60 percent of all non-hazardous construction demolition debris shall be diverted from the landfill for reuse or recycling. The contractor shall provide written certification to the COR of the type and tonnage of debris reused or recycled from the contract. All refuse and the construction demolition debris not recovered for reuse or recycling shall be disposed of at a PADEP permitted and Monroe County Municipal Waste Management Authority authorized facility. The contractor shall comply with the Monroe County Solid Waste Management Plan, PA Act 90, and all applicable licensing requirements. All waste must be transported to designated Monroe County landfills; see http://www.thewasteauthority.com for details. For questions, contact the Monroe County Municipal Waste Management Authority at (570) 643-6100. The contractor shall provide a disposal certificate or landfill weight slip to the COR for all solid waste disposed of during the performance of this contract. The COR will then provide copies of the disposal certificate or landfill weight slips with the quarterly consolidated C&D report from The Installation Planning and Maintenance Division to EB.

30. Removal Materials
The contractor must remove from the site prior to the acceptance of work by the Government, all materials not identified to remain in place, including excess paints, building materials and equipment purchased by the contractor for the execution of this project.

31. Site Preservation and Restoration
The contractor must ensure that the land resources associated under this contract be preserved in their present condition or be restored to a like condition after completion of construction. This post construction appearance will appear to be natural and not detract from the appearance of the project.

32. Trees and Shrubs Protection
The contractor must be responsible for the protection of all trees and shrubs on site. The contractor must not allow any heavy equipment, vehicular traffic or stockpiling of materials within ten feet from the drip line of any tree. The contractor must not allow any toxic materials to be stored within 100 feet (35.5 meters) of the drip line of any tree. The contractor must not nail protective devices, signs, utility boxes or other objects to trees to be retained on the site.

33. Unexploded Ordnance (UXO)
All work that is done in the UXO area must include the support and clearance by a UXO technician. In addition, all personnel entering the UXO area must have UXO recognition training. The COR will escort the contractor to the EB for UXO recognition training. The COR will provide a map of the UXO area to the contractor.

34. Water Quality
The contractor shall not pollute streams, lakes or reservoirs. All work under this contract shall be performed in such a manner that pollution will not be created in streams, surface waters or underground water located within, or adjacent to the project area. The contractor shall not spill, emit, dump or otherwise discharge any hazardous, toxic, harmful or unauthorized pollutant, substance or material, including petroleum products, cleaning agents or paints, onto the ground, into the air or into any waters or nearby storm drain. The contractor shall execute any preventative measures required to prevent any hazardous, toxic or harmful material stored or used on the project site from entering any stormwater drain. Nothing shall be allowed to spill, emit, dump or otherwise discharge any hazardous, toxic or harmful material or pollutant into any sink, toilet, drain, utility or receptacle without written permission from the EB through the COR. The contractor shall protect streams, lakes and wetlands from sediment discharges caused by his activities. The contractor shall contract the EB immediately in the event of a spill.

19. CMMC REQUIREMENTS

CONTRACTOR COMPLIANCE WITH THE CYBERSECURITY MATURITY MODEL CERTIFICATION (CMMC) LEVEL REQUIREMENTS (NOV 2025)

(a) *Definitions.* As used in this clause-

Controlled unclassified information means information the Government creates or possesses, or information an entity creates or possesses for or on behalf of the Government, that a law, regulation, or Governmentwide policy requires or permits an agency to handle using safeguarding or dissemination controls (32 CFR 2002.4(h)).

Current means—

(1) With regard to Conditional Cybersecurity Maturity Model Certification (CMMC) Status—

(i) Not older than 180 days for Conditional Level 2 (Self) assessments and Conditional Level 2 (certified third-party assessment organization (C3PAO)) assessments, with—

(A) No changes in compliance with the requirements at 32 CFR part 170 since the Conditional CMMC Status date (see 32 CFR 170.16 and 170.17); and

(B) A corresponding affirmation of continuous compliance by an affirming official (see 32 CFR 170.4); and

(ii) Not older than 180 days for Conditional Level 3 (Defense Industrial Base Cybersecurity Assessment Center (DIBCAC)) assessments, with—

(A) No changes in compliance with the requirements at 32 CFR part 170 since the Conditional CMMC Status date (see 32 CFR 170.18); and

(B) A corresponding affirmation of continuous compliance by an affirming official;

(2) With regard to Final CMMC Status—

(i) Not older than 1 year for Final Level 1 (Self), with—

(A) No changes in compliance with the requirements at 32 CFR part 170 since the Final CMMC Status date (see 32 CFR 170.15); and

(B) A corresponding affirmation of continuous compliance, not older than 1 year, by an affirming official;

(ii) Not older than 3 years for Final Level 2 (Self) assessments and Final Level 2 (C3PAO) assessments, with—

(A) No changes in compliance with the requirements at 32 CFR part 170 since the Final CMMC Status date (see 32 CFR 170.16 and 170.17); and

(B) A corresponding affirmation of continuous compliance, not older than 1 year, by an affirming official; and

(iii) Not older than 3 years for Final Level 3 (DIBCAC) assessments, with—

(A) No changes in compliance with the requirements at 32 CFR part 170 since the Final CMMC Status date (see 32 CFR 170.18); and

(B) A corresponding affirmation of continuous compliance, not older than 1 year, by an affirming official; and

(3) With regard to affirmation of continuous compliance (32 CFR 170.22), not older than 1 year with no changes in compliance with the requirements at 32 CFR part 170.

Cybersecurity Maturity Model Certification (CMMC) status means the result of meeting or exceeding the minimum required score for the corresponding assessment. The potential statuses are as follows:

(1) Final Level 1 (Self).

(2) Conditional Level 2 (Self).

(3) Final Level 2 (Self).

(4) Conditional Level 2 (C3PAO).

(5) Final Level 2 (C3PAO).

(6) Conditional Level 3 (DIBCAC).

(7) Final Level 3 (DIBCAC).

Cybersecurity Maturity Model Certification unique identifier (CMMC UID) means 10 alpha-numeric characters assigned to each CMMC assessment and reflected in the Supplier Performance Risk System (SPRS) for each contractor information system.

Federal contract information (FCI) means information, not intended for public release, that is provided by or generated for the Government under a contract to develop or deliver a product or service to the Government. It does not include information provided by the Government to the public, such as on public websites, or simple transactional information, such as information necessary to process payments.

Plan of action and milestones means a document that identifies tasks to be accomplished. It details resources required to accomplish the elements of the plan, any milestones in meeting the tasks, and scheduled completion dates for the milestones, as defined in National Institute of Standards and Technology Special Publication 800-115 (32 CFR 170.21).

(b) *Framework.* The Cybersecurity Maturity Model Certification (CMMC) is a framework for assessing a contractor's compliance with applicable information security protections (see 32 CFR part 170).

(c) *Duplication.* The CMMC assessments will not duplicate efforts from any other comparable DoD assessment, except for rare circumstances when a reassessment may be necessary, for example, when there are indications of issues with cybersecurity and/or compliance with CMMC requirements.

(d) *Requirements.* The Contractor shall—

(1)(i) Have and maintain for the duration of the contract a current CMMC status at the following CMMC level, or higher: ***CMMC Level 2 (C3PAO)*** for all information systems used in performance of the contract, task order, or delivery order that process, store, or transmit FCI or CUI; and

(ii) Consult 32 CFR 170.23 related to the flowdown of the CMMC requirements, and flow down the correct CMMC level to subcontracts and other contractual instruments;

(2) Only process, store, or transmit FCI or CUI on contractor information systems that have a CMMC status at the CMMC level required in paragraph (d)(1) of this clause, or higher;

(3) Complete on an annual basis, and maintain as current, an affirmation, by the affirming official (see 32 CFR 170.4), of continuous compliance with the requirements associated with the CMMC level required in paragraph (d)(1) of this clause in the Supplier Performance Risk System (SPRS) (*https://piee.eb.mil*) for each CMMC UID applicable to each of the contractor information systems that process, store, or transmit FCI or CUI and that are used in performance of the contract;

(4) Ensure all subcontractors and suppliers complete prior to subcontract award, and maintain on an annual basis, an affirmation, by the affirming official (see 32 CFR 170.4), of continuous compliance with the requirements associated with the CMMC level required for the subcontract or other contractual instrument for each of the subcontractor information systems that process, store, or transmit FCI or CUI and that are used in performance of the subcontract; and

(5) If the Contractor has a CMMC Status of Conditional, successfully close out a valid plan of action and milestones (32 CFR 170.21) to achieve a CMMC Status of Final.

(e) *Reporting.* The Contractor shall—

(1) Submit to the Contracting Officer—

(i) The CMMC UID(s) issued by SPRS for contractor information systems that will process, store, or transmit FCI or CUI during performance of the contract; and

(ii) Any changes in the CMMC UIDs generated in SPRS throughout the life of the contract, task order, or delivery order, if applicable;

(2) Enter into SPRS the results of a current self-assessment for each CMMC UID, not covered by a C3PAO assessment or DIBCAC assessment, applicable to each of the contractor information

systems that process, store, or transmit FCI or CUI and that are used in performance of the contract; and

(3) Complete in SPRS on an annual basis and maintain as current an affirmation of continuous compliance by the affirming official (see 32 CFR 170.4) for each self-assessment, C3PAO assessment, or DIBCAC assessment required under the contract in SPRS.

(f) *Subcontracts.* The Contractor shall—

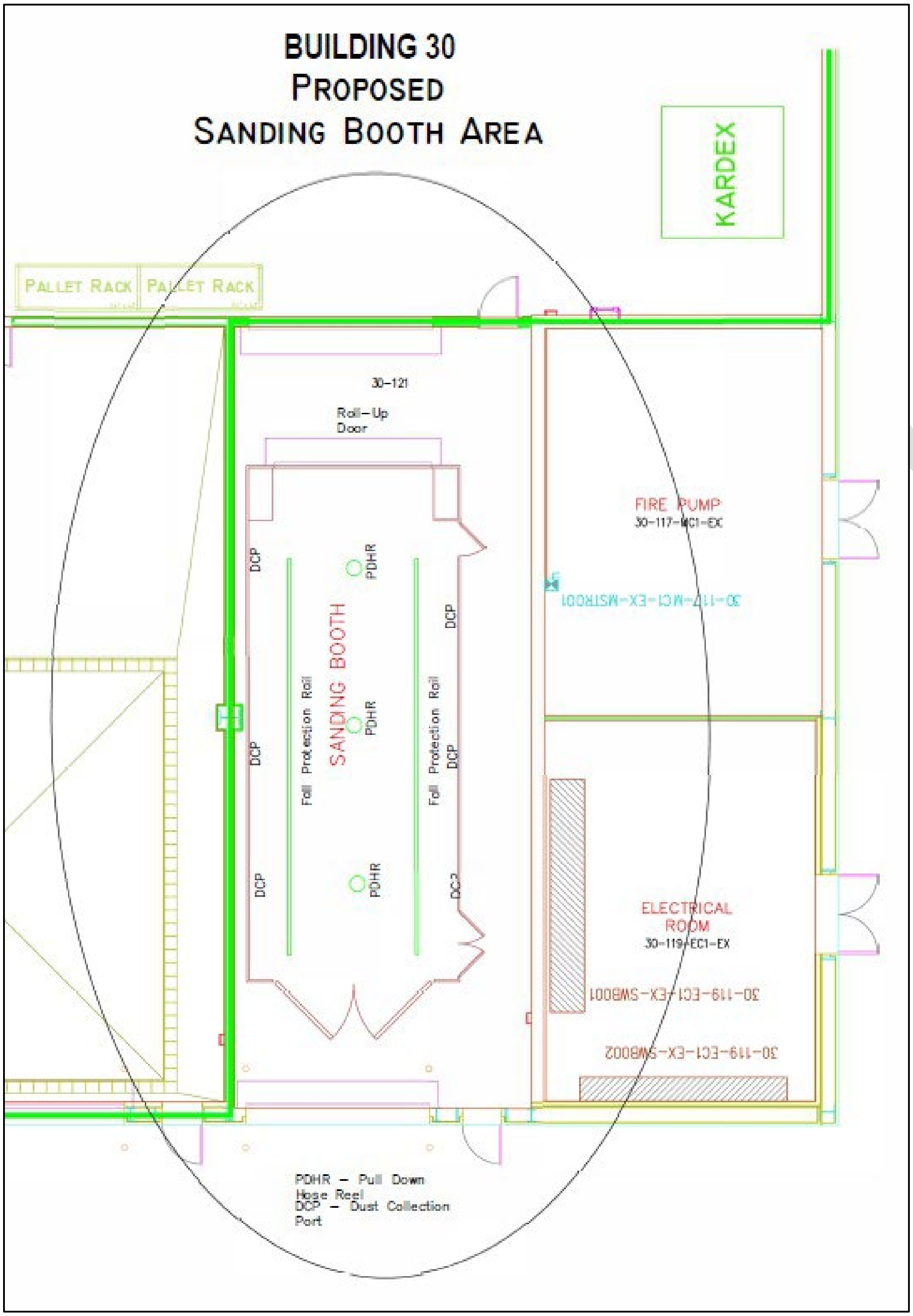
(1) Insert the substance of this clause, including this paragraph (f) and excluding paragraph (e)(1), in subcontracts and other contractual instruments, including those for the acquisition of commercial products and commercial services, excluding commercially available off-the-shelf items, if the subcontract or other contractual instrument will contain a requirement to process, store, or transmit FCI or CUI; and

(2) Prior to awarding a subcontract or other contractual instrument, ensure that the subcontractor has a current CMMC certificate or current CMMC status at the CMMC level that is appropriate for the information that is being flowed down to the subcontractor based on the requirements at 32 CFR 170.23.

(End of CMMC clause)

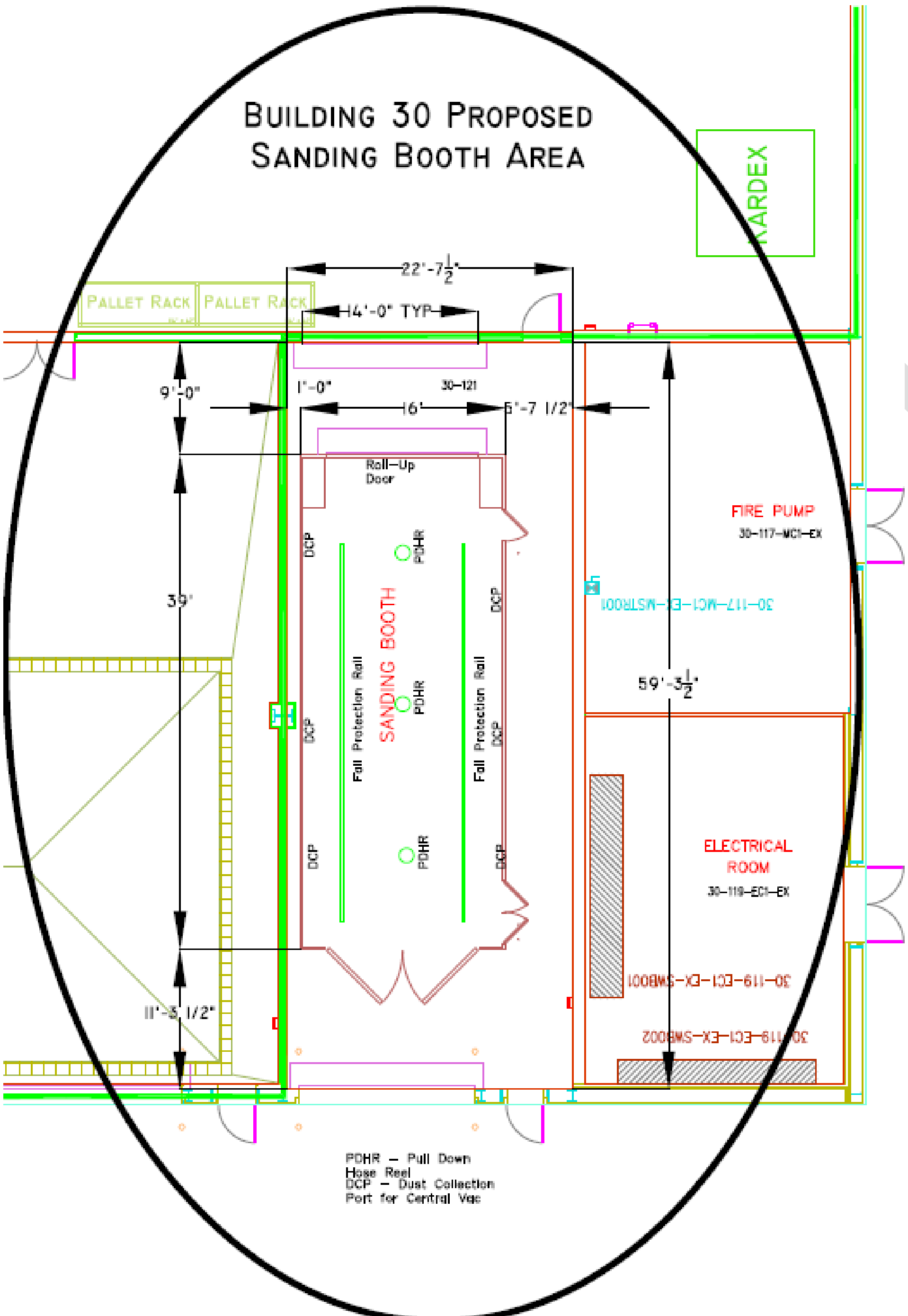
ATTACHMENT 1:

PROPOSED CONFIGURATION of Sanding Booth in Building 30



ATTACHMENT 1:

PROPOSED CONFIGURATION of Sanding Booth in Building 30 with Dimensions



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- Do Not Disseminate -

ATTACHMENT 2: Building 10A – Report from Dust Explosivity Test

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EMSL Order ID.: 362403510
Sample(s) Received: 12/31/2024
Date of Reporting: 1/16/2025
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Reported By: E. Cauley

Laboratory Report -

Explosibility of Dust Clouds

Building 10 Fiberglass Sanding Booth

Procurement of Samples and Analytical Overview:

The material for analysis (one bulk sample) arrived at EMSL Analytical, Inc. corporate laboratory in Cinnaminson, NJ on 12/31/2024. The package arrived in satisfactory condition with no evidence of damage to the contents. The data reported herein has been obtained using the following equipment and methodologies.

Methods & Equipment: OSHA ID201SG- Explosibility and Combustibility Parameters
Directive CPL 03-00-008- Combustible Dust National Emphasis Program
ASTM E1226-12- Standard Test Method for Explosibility of Dust Clouds
ASTM D6913-17 - Standard Test Methods for Particle-Si,e Distribution (Gradation) of Soils Using Sieve Analysis
X-ray fluorescence spectrometry (XRF)
XRF Standardless Analysis using Rigaku ZSX Primus wavelength-dispersive X-ray fluorescence spectrometer

20-L Siwek chamber-picture below
Mechanical Sieve system

Analyzed by: Erik Cauley — 1/9/2024
Erik Cauley
Laboratory Technician
Date

Reviewed/ Approved: Jian Hu — 1/9/2024
Jian Hu, Ph.D.
Approved Signatory
Date

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Results and Discussion:

- Go/No Go screening test was conducted using SkJ ignition source. The criterion used for a "Go" qualifier is 1-bar explosion overpressure after accounting for the influence of the 5 kJ ignition source.
- K,,values are determined using a IOkJ ignition source.

Table **1:** Combustible dust parameters for material in sample 1

Customer Sample ID	1	
Sample Description	Fibrous Glass Dust	
EMSL Sample ID	362403510-0001	
Parameter	**Value**	**Units**
Sieve (40 mesh Retained)	1.37	%wt
Sieve (40 mesh Passing)	98.63	%wt
Moisture (as received)	1.21	%wt
Combustible Material (see Note 1)	68.28	%wt
Combustible Dust (see Note 2)	67.34	%wt
Maximum Pressure (Pm-,)	8.1 ± 10%	bar
Max.Rate of Pressure Rise (dP/dt)max	586 ± 12%	barfs
Maximum Normalized Rate of Pressure Rise (Ks,)	159 ± 12%	bar m/s

According to OSHA directive CPL 03-00-008, sieving was performed on the "as received" material.
Material passing 40 mesh sieve (425 µm) was used for the analysis.

Per ASTM E1226-12 recommendations, material that contains> 5% moisture should *be* dried out prior to testing. Moisture in the dust particles raises the ignition temperature of the dust because of the heat absorbed during heating and vaporization of the moisture; therefore, the dust may present a lower explosion propensity.

The material was not dried before testing.

<u>The sample was determined to be "explosible" in dust form ("GO")</u>. Therefore, the explosion severity parameters were determined.

Note 1: % Combustible Material is defined as [100-Ash Content]% (determined as weight loss at ashing/pyrolyzing the sample for 1 hat 600"C}; defined in OSHA ID201SG-.
Note 2: % Combustible Dust=(% pass 40 mesh) * (% Combustible Material); defined in OSHA ID201SG-

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Table 2: Detailed Explosion Severity results (Kst) for material in sample 1

Customer Sample ID	Run	250 g/m'		500 g/m'		1000 g/m'		1500 g/m^3		2000 *gfm'*	
		Pm [barg]	dP/dt [bar/s]	Pm [barg]	dP/dt [bar/s]	Pm [barg]	dP/dt [barfs]	Pm [barg]	dPfdt [barfs]	Pm [barg]	dP/dt [bar/s]
1	1	7.1	342	8.1	635	8.6	669	8.3	591	5.8	393
	2		-	6.1	463	7.9	554	7.3	554		
	3	-	-	7.8	486	7.0	566	6.1	352		-

Table 3: Additional data-Particle size analysis for material in sample 1

Customer Sample ID	1
Sample Description	Fibrous Glass Dust
EMSL Sample ID	362403510-0001
Particle Size Range	**wt% in the Size Range**
2" - 1"	0.00
1'/-¾}}	0.00
3f4" -3/8"	0.00
3/8" - 4.75 mm	0.75
4.75 mm - 2.00 mm	0.12
2.00 mm - 425 µm	0.50
425 µm -75 µm	95.84
<75µm	2.80

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Table 4: Elemental composition of material in sample **1** determined by XRF analysis

Component	Concentration (wt%)	Reporting Limit (wt%)
Si02	29.988	0.033
Cao	17.687	0.010
Al203	13.533	0.059
Ti02	11.826	0.026
MgO	8.251	0.052
Br	5.860	0.004
Fe203	3.056	0.009
PbO	1.618	0.009
Cr203	1.207	0.012
Bao	1.123	0.052
ZnO	1.105	0.004
Zr02	0.971	0.023
S03	0.879	0.010
K20	0.866	0.004
Cl	0.667	0.012
Na20	0.421	0.141
P205	0.258	0.006
Bi203	0.236	0.009
Co203	0.142	0006
SrO	0.120	0.004
CuO	0.089	0.005
MnO	0.061	0.009

Light elements with atomic number <11 were not analyzed. The sample material was assumed as mixture of oxides for this analysis and the results were normalized.

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Descriptions and Definitions:

The **P,,,,.** is the average ma,imum pressure (above pressure in the vessel at the time of ignition) reached during the course of a deflagration for the optimum concentration of the dust tested. Pmax is determined by a series of tests over a large range of concentrations

The **(dP/dt)m,.** is the average maximum value for the rate of pressure increase per unit time reached during the course of a deflagration for the optimum concentration of dust tested. It is determined by a series of tests over a large range of concentrations.

The **Deflagration Index, K,.** is the average maximum dP/dt normali2ed to a 1.0-m' volume. It is measured at the optimum dust concentration. Kst is defined in accordance with the following cubic relationship:
Ket�(dP/dt)ma, x $V^{1}1^{3}$

% Combustible Material is defined the same way as% Ash Content (determined by ashing/pyrolyzing the sample for 1 hat 600°[).
% Combustible Dust�(% pass 40 mesh)•(% Combustible Material)

The following is a partial listing of definitions based on NFPA standards and 29 CFR 1910.399, the definitions provision of Subpart 5-Electrical, that relate to combustible dust.

A. ***Class** II **locations.*** Class II locations are those that are hazardous because of the presence of combustible dust. The following are ClciSS II locations where the combustible dust dtmospheres are present:

Group E. Atmospheres containing combustible metal dusts, including aluminum, magnesium, and their commercial alloys, and other combustible dusts whose particle size, abrasiveness, and conductivity present similar hazards in the use of electrical equipment.

Group F. Atmospheres containing combustible carbonaceous dusts that have mare than 8 percent total entrapped volatiles (see ASTM D 3175, *Standard Test Method for Volatile Matter in the Analysis Sample of Coal and Coke,* for coal and coke dusts) or that have been sensitized by other materials so that they present an explosion hazard. Coal, carbon black, charcoal, and coke dusts are examples of carbonaceous dusts.

Group G. Atmosphere containing other combustible dusts, including flour, grain, wood flour, plastic and chemicals.

B. ***Combustible dust.*** A combustible particulate solid that presents a fire or deflagration hazard when suspended in air or some other oxidizing medium over a range of concentrations, regardless of particle size or shape.

C. ***Combustible Particulate Solid.*** Any combustible solid material composed of distinct particles or pieces, regardless of size, shape, or chemical composition.

D, ***Hybrid Mixture.*** A mi,ture of a flammable gas with either a combustible dust or a combustible mist.

E. ***Deflagration.*** Propagation of a combustion zone at a speed that is less than the speed of sound in the unreacted medium.

F. ***Deflagration Iso/atian.*** Method employing equipment and procedures that interrupts the propagation of a deflagration of a flame front, past a predetermined point.

G, ***Deflagratian Suppression,*** The technique of detecting and arresting combustion in a confined space while the combustion is still in its incipient stage, thus preventing the development of pressures that could result in an explosion.

H. ***Detonation.*** Propagation of a combustion zone at a velocity that is greater than the speed of sound in the unreacted medium.

Dust-ignitianproaf. Equipment enclosed in a manner that excludes dusts and does not permit arcs, sparks, or heat otherwise generated or liberated inside of the enclosure to cause ignition of exterior accumulations or atmospheric suspensions of a specified dust on or in the vicin1ty of the enclosure.

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J. ***Dusttight.*** Enclosures constructed so that dust will not enter under specified test conditions.

K. ***Explosion.*** The bursting or rupture of an enclosure or a container due to the development of internal pressure from deflagration.

L. ***Minimum Explosible Concentration (MEC).*** The minimum concentration of combustible dust suspended in air, measured in mass per unit volume that will support a deflagration.

Following conditions may also indicate that a potential dust deflagration, other fire, or explosion hazard exists:

a) Plant History of Fires: The plant has a history of fires involving combustible dusts.

b) **Material Safety D.ita Sheets (MSDS):** The MSDS may indicate that a particular dust is combustible and can cause explosions, deflagrations, or other fires. However, do not use MSDSs as a sole source of information because this information is often excluded from MSDSs.

c) **Dust Accumulations:** Annex D of NFPA 654 contains guidance on dust layer characterization and precautions. It indicates that immediate cleaning is warranted whenever a dust layer of 1/32- inch thickness accumulates over a surface area of at least 5% of the floor area of the facility or any given room. The 5% factor should not be used if the floor area exceeds 20,000 ft2, in which case a 1,000 ft2 L3yer of dust is the upper limit. Accumulations on overhead beams, joists, ducts, the tops of equipment, cind other surfaces should be included when determining the dust coverage area. Even vertical surfaces should be included if the dust is adhering to them. Rough calculations shaw that the available surface area of bar joists is approximately 5 % of the floor area and the equivalent surface dre;:i for steel beams can he as high as 10%. The material in Annex Dis dn ide;:illzed cippr□ach bdsed an certain assumptions, including uniformity of the dust layer covering the surfaces, a bulk density of 75 lb/ ft3, a dust concentration of 0.35 oz/ ft 3, and a dust cloud height of 10 ft. Additionally, FM Data Sheet 7-76 contains a formula to determine the dust thickness that may create an explosion hazard in a room, when some of these variables differ.

d) Observe areas of the plant for accumulations of hazardous levels of dust (for example, greater than 1/32 of an inch, which is approximately equal to the thickness of a typical paper clip). Likely areas of dust accumulations within a plant are:

- Structural members
- Conduits and pipe racks
- Cable trays
- Floors
- Above ceiling
- On and around equipment (leaks around dust collectors and ductwork)

e) If potential combustible dust hazards are found, dust samples must be safely collected.

- "High spaces" such as roof beams, open web beams, tops of pipes and ductwork, and other horizontal surfaces located as high in the overhead as possible. Note: These are the preferred locations; however, if a means of safe access is not available, sample(s) should not be collected
- Equipment and floors where dust has accumulated
- The interior (i.e., bins and/or bags) of a dust collector
- Within ductwork

(From **OSHA Directive No. CPL-03-00-008)**

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Additional Reference Information:

Hazardous (classified) locations.

This section covers the requirements far electric equipment and wiring in locations which are classified depending on the properties of the flammable vapors, liquids or gases, or combustible dusts or fibers which may be present therein and the likelihood that a flammable or combustible concentration or quantity is present. Hazardous (classified) locations may be found in occupancies such as, but not limited to, the following: aircraft hangars, gasoline dispensing and service stations, bulk storage plants for gasoline or other volatile flammable liquids, paint-finishing process plants, health care facilities, agricultural or other facilities where excessive combustible dusts may be present, marinas, boat yards, and petroleum and chemical processing plants. Each room, section or area shall be considered individually in determining its classification. These hazardous (classified) locations are assigned six designations as follows:
Class I, Division 1
Class I, Division 2
Class II, Division 1
Class II, Division 2
Class III, Division 1
Class III, Division 2

Class I locations. Class I locations are those in which flammable gases or vapors are or may be present in the air in quantities sufficient to produce explosive or ignitable mixtures. Class I locations include the following:

Iii *Class I, Division 1.* A Class I, Division 1 location is a location: (a) in which hazardous concentrations of flammable gases or vapors may exist under normal operating conditions; or (b) in which hazardous concentrations of such gases or vapors may exist frequently because of repair or maintenance operations or because of leakage; or (c) in which breakdown or faulty operation of equipment or processes might release hazardous concentrations of flammable gases or vapors, and might also ca use simultaneous failure of electric equipment.

Nate: This classification usually includes locations where volatile flammable liquids or liquefied flammable gases are transferred from one container to another; interiors of spray booths and areas in the vicinity of spraying and painting operations where volatile flammable solvents are used; locations containing open tanks or vats of volatile flammable liquids; drying rooms or compartments for the evaporation of flammable solvents; locations containing fat and oil extraction equipment using volatile flammable solvents; portions of cleaning and dyeing plants where flammable liquids are used; gas generator rooms and other portions of gas manufacturing plants where flammable gas may escape; inadequately ventilated pump rooms for flammable gas or for volatile flammable liquids; the interiors of refrigerators and freezers in which volatile flammable materials are stored in open, lightly stoppered, or easily ruptured containers; and all other locations where ignitable concentrations of flammable vapors or gases are likely to occur in the course of normal operations.

(ii) Class I, Division 2. A Class I, Division 2 location is a location: (a) in which volatile flammable liquids or flammable gases are handled, processed, or used, but in which the hazardous liquids, vapors, or gases will normally be confined within closed containers or closed systems from which they can escape only in case of accidental rupture or breakdown of such containers or systems, or in case of abnormal operation of equipment; or lb) in which hazardous concentrations of
gases or vapors are normally prevented by positive mechanical ventilation, and which might become hazardous through failure or abnormal operations of the ventilating equipment; or (c) that is adjacent to a Class I, Division 1 location, and to which hazardous concentrations of gases or vapors might occasionally be communicated unless such communication is prevented by adequate positive-pressure ventilation from a source of clean air, and effective safeguards against ventilation failure are provided.

Note: This classification usually includes locations where volatile flammable liquids or flammable gases or vapors are used, but which would become hazardous only in case of an accident or of some unusual operating condition. The quantity of flammable material that might escape in case of accident, the adequacy of ventilating equipment, the total area involved, and the record of the industry or business with respect to explosions or fires are all factors that merit consideration in determining the classification and e:xtent of each location.

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Piping without valves, checks, meters, and similar devices would not ordinarily introduce a hazardous condition even though used for flammable liquids or gases. Locations used for the storage of flammable liquids or a liquefied or compressed gases in sealed containers would not normally be considered hazardous unless also subject to other hazardous conditions.

Electrical conduits and their associated enclosures separated from process fluids by a single seal or barrier are classed as a Division 2 location if the outside of the conduit and enclosures is a non-hazardous location.

Class II **locations:** Class II locations are those that are hazardous because of the presence of combustible dust. Class II locations include the following:

Iii *Class* II, *Division 1*. A Class II, Division 1 location is a location: (a) In which combustible dust is or may be in suspension in the air under normal operating conditions, in quantities sufficient to produce explosive or ignitable mixtures; or (bl where mechanical failure or abnormal operation of machinery or equipment might cause such explosive or ignitable mixtures to be produced, and might also provide a source of ignition through simultaneous failure of electric equipment, operation of protection devices, or from other causes, or le) in which combustible dusts of an electrically conductive nature may be present.

Note: This classification may include areas of grain handling and processing plants, starch plants, sugar-pulverizing plants, malting plants, hay-grinding plants, coal pulverizing plants, areas where metal dusts and powders are produced or processed, and other similar locations which contain dust producing machinery and equipment (except where the equipment is dust-tight or vented to the outside). These areas would have combustible dust in the air, under normal operating conditions, in quantities sufficient to produce explosive or ignitable mixtures. Combustible dusts which are electrically nonconductive include dusts produced in the handling and processing of grain and grain products, pulverized sugar and cocoa, dried egg and milk powders, pulverized spices, starch and pastes, potato and wood flour, oil meal from beans and seed, dried hay, and other organic materials which may produce combustible dusts when processed or handled. Dusts containing magnesium or aluminum are particularly hazardous and the use of extreme caution is necessary to avoid ignition and explosion.

Iii) *Closs* II, *Division 2*. A Class II, Division 2 location is a location in which: (a) combustible dust will not normally be in suspension in the air in quantities sufficient to produce explosive or ignitable mixtures, and dust accumulations are normally insufficient to interfere with the normal operation of electrical equipment or other apparatus; or (b) dust may be
in suspension in the air as a result of infrequent malfunctioning of handling or processing equipment, and dust accumulations resulting there from may be ignitable by abnormal operation or failure of electrical equipment or other apparatus.

Note: This classification includes locations where dangerous concentrations of suspended dust would not be likely but where dust accumulations might form on or in the vicinity of electric equipment. These areas may contain equipment from which appreciable quantities of dust would escape under abnormal operating conditions or be adjacent to a Class II Division 1 location, as described above, into which an explosive or ignitable concentration of dust may be put into suspension under abnormal operating conditions.

Class III **locations.** Class III locations are those that are hazardous because of the presence of easily ignitable fibers or flyings but in which such fibers or flyings are not likely to be in suspension in the air in quantities sufficient to produce ignitable mixtures. Class III locations include the following:

Iii *Class III, Division 1*. **A** Class III, Division 1 location is a location in which easily ignitable fibers or materials producing combustible flyings are handled, manufactured, or used.
Note: Such locations usually include some parts of rayon, cotton, and other textile mills; combustible fiber manufacturing and processing plants; cotton gins and cotton-seed mills; flax-processing plants; clothing manufacturing plants; woodworking plants, and establishments; and industries involving similar hazardous processes or conditions.
Easily ignitable fibers and flyings include rayon, cotton (including cotton linters and cotton waste), sisal or henequen, istle, jute, hemp, tow, cocoa fiber, oakum, baled waste kapok, Spanish moss, excelsior, and other materials of similar nature.

(iii *Class III, Division 2*. A Class III, Division 2 location is a location in which easily ignitable fibers are stored or handled, except in process of manufacture.

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Appendix E, Sub section 5, Maximum Normalized Rate of Pressure Rise

Dust Explosion Class	***Kst (bar.mis)***	***Characteristic***
St 0	0	No Explosion
St 1	>O and <200	Weak Explosion
St 2	>200 and <300	Strong Explosion
St 3	>300	Very Strong Explosion

Table from OSHA CPL 03-00-008: Explosion class determination from the maximum normalized rate of explosion.

(from **PART 1910- Occupational Safety and Health Standards**
Subpart S- Electrical Design Safety Standards for Electrical Systems)

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Important Terms Conditions and Limitations:

Sample Retention: Samples analyzed by EMSL will be retained for 60 days after analysis date. Storage beyond this period is available for a fee with written request prior to the initial 30 day period. Samples containing hazardous/toxic substances which require special handling may be returned to the client immediately. EMSL reserves the right to charge a sample disposal or return shipping fee.

Change Orders and Cancellation: All changes in the scope of work or turnaround time requested by the client after sample acceptance must be made in writing and confirmed in writing by EMSL. If requested changes result in a change in cost the client must accept payment responsibility. In the event work is cancelled by a client, EMSL will complete worl, in progress and invoice for work completed to the point of cancellation notice. EMSL is not responsible for holding times that are exceeded due to such changes.

Warranty: EMSL warrants to its clients that all services provided hereunder shall be performed in accordance with established and recognized analytical testing procedures, when available. The foregoing express warranty is exclusive and is given in lieu of all other warranties, expressed or implied. EMSL disclaims any other warranties, express or implied, including a warranty of fitness for particular purpose and warranty of merchantability.

Limits of Liability: In no event shall EMSL be liable for indirect, special, consequential, or incidental damages, including, but not limited to, damages for loss of profit or goodwill regardless of the negligence (either sole or concurrent) of EMSL and whether EMSL has been informed of the possibility of such damages, arising out of or in connection with EMSL's services thereunder or the delivery} use, reliance upon or interpretation of test results by client or any third party. We accept no legal responsibility for the purposes for which the client uses the test results. EMSL will not be held responsible for the improper selection of sampling devices even if we supply the device to the user. The user of the sampling device has the sole responsibility to select the proper sampler and sampling conditions to ensure that a valid sample is taken for analysis. Any resampling performed will be at the sole discretion of EMSL, the cost of which shall be limited to the reasonable value of the original sample delivery group (SDG) samples. In no event shall EMSL be liable to a client or any third party, whether based upon theories of tort, contract or any other legal or equitable theory, in excess of the amount paid to EMSL by client thereunder.

The data and other information contained in this report, as well as any accompanying documents, represent only the samples analyzed. They are reported upon the condition that they are not to be reproduced wholly or in part for advertising or other purposes without the written approval from the laboratory.